

Modelling COVID-19: Mathematical Modelling Approaches to Study Infectious Diseases

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1 Introduction

1.1 Introduction

The aim of this dissertation is to develop, analyse, and computationally simulate infectious diseases by applying Ordinary Differential Equations (ODEs) and machine learning based predictive tools to COVID-19, to evaluate disease spread.

Infectious diseases remain among the most significant threats to global health [1]. The COVID-19 pandemic was a clear demonstration of how quickly a pathogen can spread through a population. COVID-19, caused by the SARS-CoV-2 virus, is a highly transmissible respiratory infection characterised by fever, dry cough, and fatigue, often with pulmonary involvement [2]. A central feature of SARS-CoV-2 is its ability to spread through human to human transmission, which meant that widespread circulation even in short time periods was possible [33, 5]. In addition, SARS-CoV-2 exhibits a relatively short serial interval compared with many respiratory infections, which allows new infections to occur in quick succession and contributes to the rapid escalation of cases [37]. Given the contagious nature of COVID-19, diverse transmission routes, and short serial interval, this dissertation focuses on understanding how this highly transmissible pathogen spreads through a population by developing and analysing epidemic models.

The rapid spread of COVID-19 highlighted the need for reliable tools to predict epidemic behaviour. Mathematical modelling plays a key role in this, as it allows us to test our understanding of disease dynamics while generating predictions that guide interventions [26]. Importantly, by using mathematical models, we can investigate epidemics without the cost, time, or logistical challenges of real world data collection [28]. By building these models and computational simulations, we are able to explore how infection rates, recovery rates, behavioural changes, and vaccination strategies influence overall epidemic outcomes. In this dissertation, these modelling tools are applied to SIS, SIR, SEIR, and COVID-19 specific models, specifically to explore how both classical mathematical theories and modern computations can contribute to understanding and controlling the spread of infectious disease.

1.2 Mathematical Models in Epidemiology

Mathematical modelling of infectious disease dates back as early as the work of Kermack and McKendrick in 1927, whose SIR framework established the foundation for modern epidemic theory [29]. Compartmental mathematical models split populations into distinct disease states. The core states in these models are susceptible (S), exposed (E), infected (I), and recovered (R), with transitions between each one of these states governed by differential equations that capture disease transmission dynamics [8]. These models provide a framework for understanding how diseases spread through populations and for evaluating the impact of control measures.

During the COVID-19 pandemic, compartmental models were deployed by research groups and public health agencies globally to inform critical policy decisions [56]. These models helped in understanding epidemic dynamics and identifying key parameters such as the basic reproduction number R_0 , which determines whether an outbreak will grow or decline. A value of $R_0 > 1$ indicates that each infected individual will infect more than one other person, leading to exponential growth of the epidemic, while $R_0 < 1$ suggests the epidemic will eventually die out [8].

However, the pandemic also revealed limitations of traditional compartmental models [24]. Standard epidemic models often struggle to represent contact patterns across different population groups due to the assumption of homogeneous mixing [13]. These limitations have increased interest in complementary approaches, including machine learning methods [47], which are able to identify patterns in epidemic data without requiring a researcher to define the biological or behavioural processes.

1.3 Outline

This project will explore the dynamics of COVID-19 through the use of compartmental models and machine learning approaches. These include the basic SIS model, the SIR model, SEIR type models that will incorporate a disease's latent period, hybrid models that will combine machine learning's predictive capacity with compartmental models' mechanistic structure, and finally, extending the analysis to neural networks to learn the relationship between different data points. This project will analyse these models numerically and analytically in order to visualise disease spread as a way of determining whether epidemic or endemic behaviour occurs.

Moreover, this dissertation will explore and develop extended compartmental models to better capture COVID-19 disease dynamics. In order to improve the predictive accuracy of compartmental models, additional compartments such as

vaccination, waning immunity, and disease induced deaths are incorporated, allowing for a more realistic representation of how interventions and behavioural factors influence the course of an epidemic. By combining these extended frameworks with machine learning techniques, this work aims to demonstrate how traditional epidemiological models can be enhanced to provide more robust and data driven insights into the spread and control of infectious diseases.

Finally, this dissertation will scrutinise control and eradication strategies using the proposed models. By simulating the effects of interventions such as vaccinations, social distancing measures, and quarantine policies, the models developed throughout this work can be used to evaluate which strategies are most effective in reducing transmission and ultimately controlling or eradicating the disease. The analysis will provide insight into the conditions under which COVID-19 can transition from epidemic to endemic behaviour, and the extent to which different intervention strategies can influence these outcomes.

2 Mathematical Modelling of Epidemics

2.1 Introduction

This section introduces fundamental compartmental models used in epidemiology, that form the basis for COVID-19 analysis.

2.2 SIS Epidemics

2.2.1 Model Formulation

In a simple epidemic, the population consists of two groups: susceptibles S , those who can catch the disease, and infectives I , individuals who are currently infected and can transmit the disease. Although COVID-19 is not an SIS disease, the SIS model offers a useful approximation for its early dynamics, particularly before vaccinations were established.

In a simple epidemic, we assume that a susceptible, once infected, becomes infectious immediately and later becomes susceptible again [8]:

$$S \rightarrow I \rightarrow S$$

This is an approximation to the initial stages of many diseases. This idea of entering back into the susceptible compartment is a key dynamic observed during COVID-19, as many individuals experienced waning immunity [48] which meant that once they became, infected they were not guaranteed immunity.

We can also assume that the total population size is a constant N for the epidemic, so that

$$S(\tau) + I(\tau) = N$$

where $S(\tau)$ and $I(\tau)$ are the numbers of susceptible and infectious individuals at time τ [8].

Differential equations are used to represent the rate at which an individual moves through the system. The movement for S and I are given by

$$\frac{dS}{d\tau} = -f(S, I), \quad \frac{dI}{d\tau} = f(S, I),$$

where $f(S, I)$ is the rate at which new infections occur. In general, $f(S, I)$ is an increasing function of both S and I . The simplest model is:

$$f(S, I) = \beta IS,$$

where the parameter $\beta > 0$ is the infectious transmission rate measuring how effectively the disease spreads between susceptible and infectious individuals.

We can also introduce the idea that an infectious individual I can recover but move back to the susceptible compartment, similar to most infections. We can denote this by $g(I)$ *recovery from the disease* it can be written as

$$g(I) = \gamma I$$

where $\gamma > 0$ is the *recovery rate* from an infection.

The SIS model can be written as a system of ordinary differential equations (ODEs):

$$\begin{aligned}\frac{dS}{d\tau} &= -f(S, I) + \gamma I = -\beta \frac{SI}{N} + \gamma I, \\ \frac{dI}{d\tau} &= f(S, I) - \gamma I = \beta \frac{SI}{N} - \gamma I,\end{aligned}$$

where the term $\beta \frac{SI}{N}$ represents the flow from the susceptible group to the infectious group and γI is the rate of recovery of the infection [7].

Rearranging the basic epidemic equation, we get $S = N - I$, the system can be reduced to a single equation for $I(\tau)$:

$$\frac{dI}{d\tau} = \beta \frac{(N - I)I}{N} - \gamma I = I \left(\beta \frac{N - I}{N} - \gamma \right).$$

2.2.2 The basic reproduction number

In epidemiological modelling, a fundamental question is whether an infectious disease can survive within a fully susceptible population. The value used to show this is the *basic reproduction number*, denoted by R_0 [41]. The basic reproduction number R_0 was important during the COVID-19 pandemic, as it informed government prevention programmes [3].

To determine when an SIS infection can establish itself in a population, we analyse the equilibrium points from the reduced equation. Setting $\frac{dI}{d\tau} = 0$, we obtain:

$$I \left(\beta \frac{N - I}{N} - \gamma \right) = 0.$$

This gives either the disease free equilibrium $I^* = 0$, or the endemic equilibrium satisfying

$$\beta \frac{N - I^*}{N} - \gamma = 0.$$

Rearranging gives $\beta(N - I^*) = \gamma N$, and solving for I^* gives:

$$I^* = N \left(1 - \frac{\gamma}{\beta} \right).$$

For this endemic equilibrium to be meaningful, we require $I^* > 0$, which holds if and only if $\beta > \gamma$, or $\frac{\beta}{\gamma} > 1$. This ratio represents the average number of secondary infections generated by a single infectious individual in a susceptible population. This means the definition of the basic reproduction number is:

$$R_0 = \frac{\beta}{\gamma}.$$

The endemic equilibrium can then be expressed as

$$I^* = N \left(1 - \frac{1}{R_0} \right).$$

2.2.3 Numerical simulation of the SIS model

The purpose of this section is to illustrate how the SIS model behaves using numerical simulations, showing how different values of β (*infection rate*) and γ (*recovery rate*) influence the spread of infections. Since the reproduction ratio is:

$$R_0 = \frac{\beta}{\gamma}$$

and determines the contagiousness or transmissibility of an infection [14]. We simulate the SIS model with the parameters $R_0 < 1$ and $R_0 > 1$, which allows us to visualise when the disease dies out ($R_0 < 1$) and when the infection remains endemic in the population ($R_0 > 1$) [8].

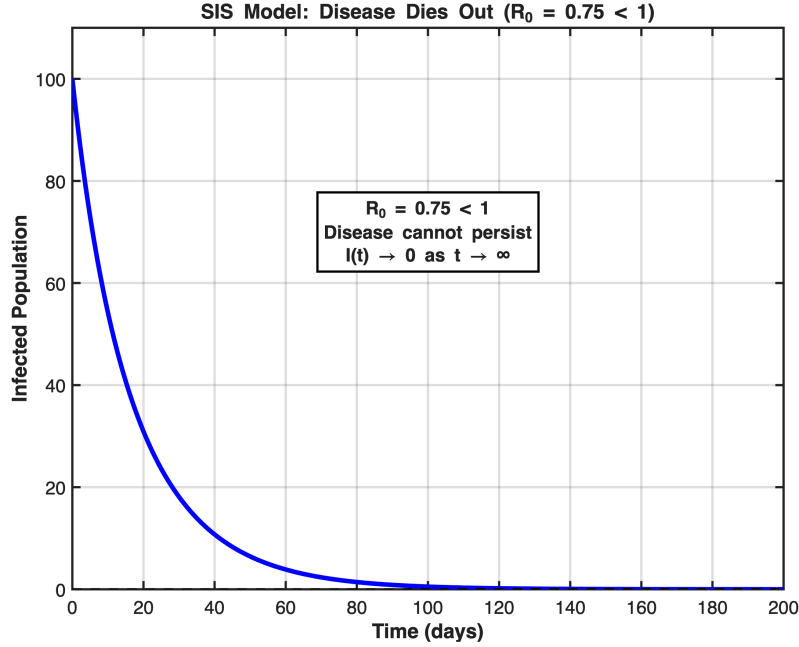


Figure 1: SIS model simulation with $R_0 = 0.75 < 1$. The infected population decays exponentially to zero, demonstrating stability of the disease free equilibrium.

Figure 1 shows that when $R_0 = 0.75 < 1$, the infected population decays exponentially to zero over time, demonstrating that the disease free equilibrium is

stable. This behaviour comes from the SIS model dynamics, since $R_0 < 1$ each infected individual transmits the disease to fewer than one new person on average. This means that the infected compartment gets smaller over time until the disease can no longer sustain itself within the population. During COVID-19, achieving $R_0 < 1$ was the goal of public health interventions such as lockdowns, mask mandates, and social distancing measures. These interventions worked by reducing the transmission rate β , bringing the value of R_0 below the threshold of 1.

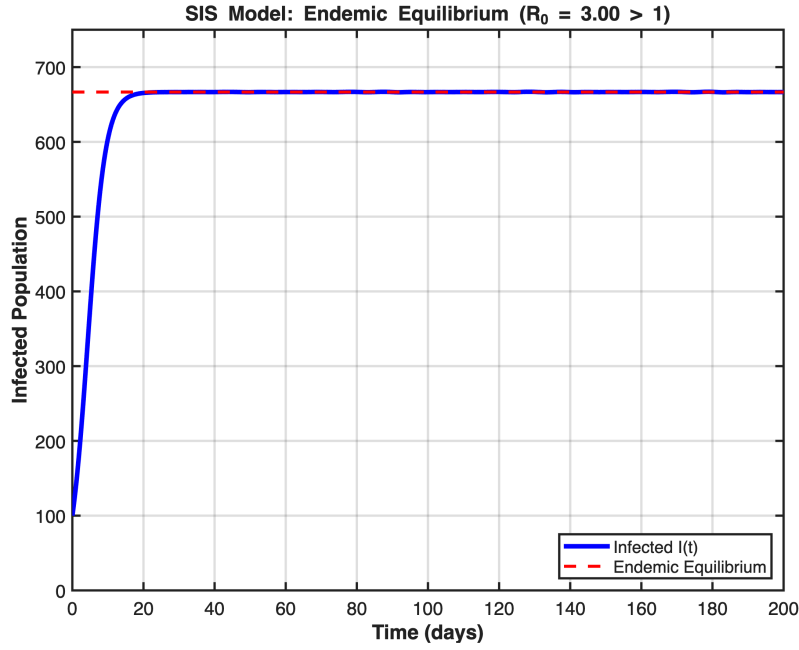


Figure 2: SIS model simulation with $R_0 = 3.0 > 1$. The infected population converges to the endemic equilibrium, where the disease is constant in the population.

Figure 2 shows that when $R_0 = 3 > 1$, the infected population increases rapidly before converging to the endemic equilibrium over time, confirming that the disease is present within the population. This behaviour comes from the SIS model dynamics, since $R_0 > 1$, each infected individual typically transmits the disease to more than one new person. This means that the infected compartment gets larger over time until it stabilises at the endemic equilibrium value $I^* = N(1 - \frac{1}{R_0})$. For COVID-19, early estimates placed R_0 between 2 and 3 [3].

2.3 SIR Epidemics

2.3.1 Model Formulation

In an SIR epidemic model, individuals leaving the infective class no longer play a role in the disease, as they are assumed to be recovered with immunity, dead, or removed by an isolation policy. Such policies were widely implemented during the COVID-19 pandemic to limit the spread of infection [8, 2]. However, COVID-19 does not strictly conform to a typical SIR disease as immunity was not always permanent for an individual and reinfection was possible [45].

Despite this limitation, the SIR model remains an important continuation of the SIS framework. It permanently removes an idea of an infected individual moving back into the susceptible population. We can represent the model diagrammatically by

$$S \rightarrow I \rightarrow R$$

During this explanation of an SIR epidemic, we will assume that the duration of the epidemic is short compared to the lifetime of its hosts. This means that we can neglect births and deaths. The population is therefore closed, of size N , so it can be modelled as

$$N = S(\tau) + I(\tau) + R(\tau).$$

We model the movement between the classes as the following set of differential equations:

$$\frac{dS}{d\tau} = -\beta \frac{SI}{N}, \quad \frac{dI}{d\tau} = \beta \frac{SI}{N} - \gamma I, \quad \frac{dR}{d\tau} = \gamma I.$$

2.3.2 Numerical Simulations of the SIR Model

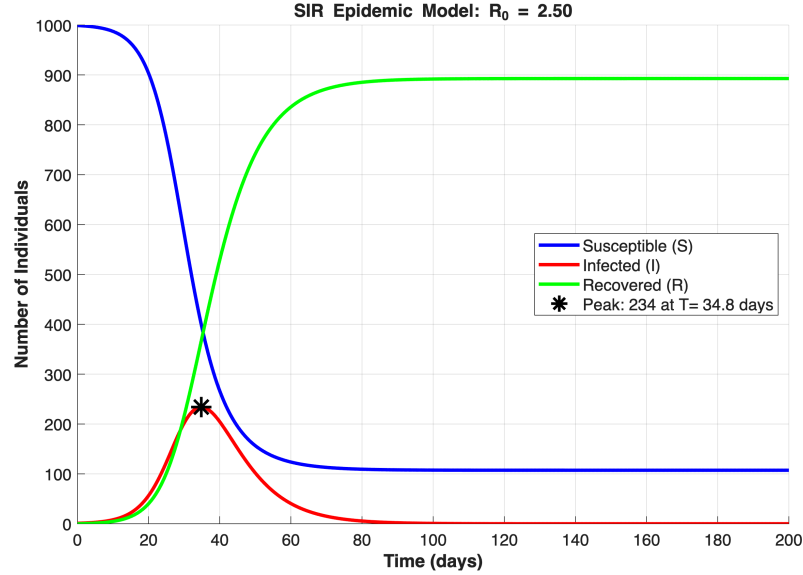


Figure 3: SIR model showing the dynamics of susceptible (S), infected (I), and recovered (R) populations. The epidemic peaks at $T = 34.8$ days with 234 infected individuals, before declining as the susceptible population is reduced and recovered approaches its final population.

Figure 3 shows a model of the SIR epidemic with $R_0 = 2.50 > 1$. It shows the susceptible, infected, and recovered populations over a period of 200 days. The infected population rises from the initial conditions, reaching a peak of 234 infected individuals at $T=34.8$ days, shown by the asterisk on the infected curve, before declining as the number of susceptible individuals enters the recovered compartment. This behaviour is a common characteristic of SIR model dynamics, since $R_0 > 1$, each infected individual transmits the disease to more than one susceptible person on average, thus driving the initial growth of infection. However, in the SIR model, individuals gain permanent immunity from the infection. This means once an individual enters the recovered compartment, they cannot return to the susceptible compartment. The parameters of this simulation are chosen to replicate early COVID-19 disease dynamics for the basic reproduction number R_0 .

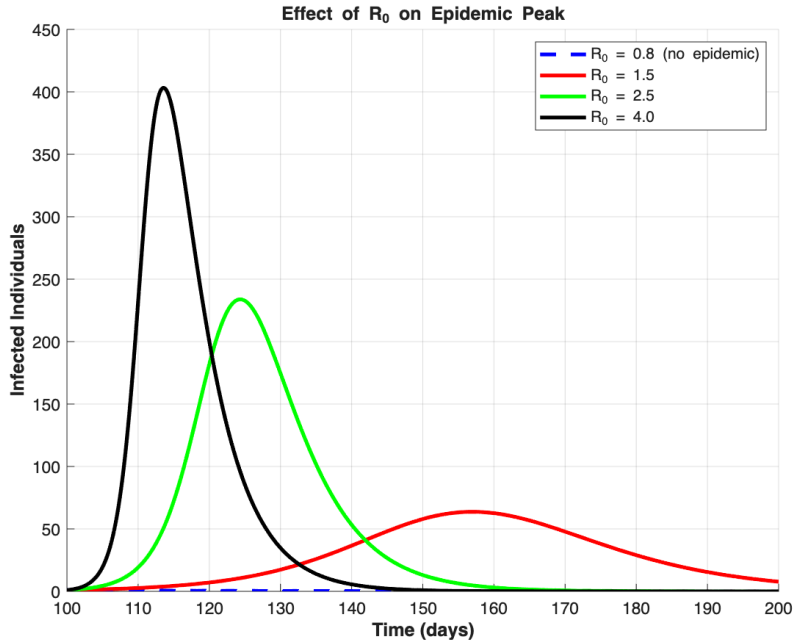


Figure 4: Impact of the basic reproduction number R_0 on SIR epidemic dynamics. The following graph shows infected population curves for different R_0 values. Demonstrating that when $R_0 < 1$, the epidemic dies out, while $R_0 > 1$ leads to epidemic spread, with the level of seriousness increasing with R_0 .

Figure 4 demonstrates the impact of the basic reproduction number R_0 on SIR epidemic dynamics by comparing infected population curves across four different values: these were $R_0 = 0.8, 1.5, 2.5$ and 4.0 . When $R_0 = 0.8$, each infected individual transmits the disease to fewer than one new person on average, meaning that the infected population fails to grow, and the epidemic dies out, showing that the disease free equilibrium is stable. For the remaining values, an epidemic does occur, but the dynamics are all distinctly different. Higher values of R_0 produce sharper, earlier epidemic peaks with a greater number of infected individuals at one time, while lower values above the threshold produce broader, flatter curves that peak later. This reinforces that a larger R_0 drives faster reduction of the susceptible population, compressing the epidemic into a shorter but more intense period. These results highlight the critical role R_0 plays in determining both the timing and severity of an outbreak.

2.4 SEIR Epidemics

2.4.1 Model Formulation

The Basic SEIR model is a closed population, compartmental model. The model has four compartments, which are S (susceptible), E (exposed), I (infectious), and R (recovered). This gives the total population as $N = S + E + I + R$, which is the total number of people in a given population [23]. The fundamental idea of an SEIR epidemic is that each member of a population moves into four compartments as time progresses. We can represent this movement diagrammatically by

$$S \rightarrow E \rightarrow I \rightarrow R.$$

The movement between these compartments can be modelled by the following system of ordinary differential equations:

$$\frac{dS}{d\tau} = -\beta \frac{SI}{N}, \frac{dE}{d\tau} = \beta \frac{SI}{N} - \delta E, \frac{dI}{d\tau} = \delta E - \gamma I, \frac{dR}{d\tau} = \gamma I.$$

Where δ is the rate of progression from exposed to infectious, and β and γ keep their definitions from the SIS and SIR model as the transmission and recovery rates of the model.

2.4.2 The Role of the Latent Period

The key difference between the SIR and SEIR models is the inclusion of the exposed class E . In the SIR model, a susceptible individual who contracts the disease becomes instantly infectious [8]. In the real world, epidemics have an incubation period where the individual is infected but not yet able to transmit the disease to anyone else in the population. This is called the latent or exposed period of an infection. The rate at which individuals can move between these compartments is governed by δ . On average, an individual spends $\frac{1}{\delta}$ days in the exposed compartment before becoming infectious. For COVID-19, the incubation period was estimated to be approximately 5 days [42], which meant that the value of δ was approximately 0.2.

The effect of the latent period on epidemic dynamic curves is a delay and a slightly reduced epidemic peak compared to the SIR model. This occurs because new infections must pass through the exposed compartment before entering the infectious compartment, thus slowing the rate at which the infectious population grows. As the latent period $\frac{1}{\delta}$ increases, this delay becomes more pronounced, resulting in a broader, flatter epidemic curve.

Despite the introduction of an additional compartment, the basic reproduction number for the SEIR model remains $R_0 = \frac{\beta}{\gamma}$, which is the same as that of the SIS and SIR models. This is because the exposed class introduces a delay but does not alter the number of new infections produced by a single infectious individual.

2.4.3 Numerical Simulations of the SEIR Model

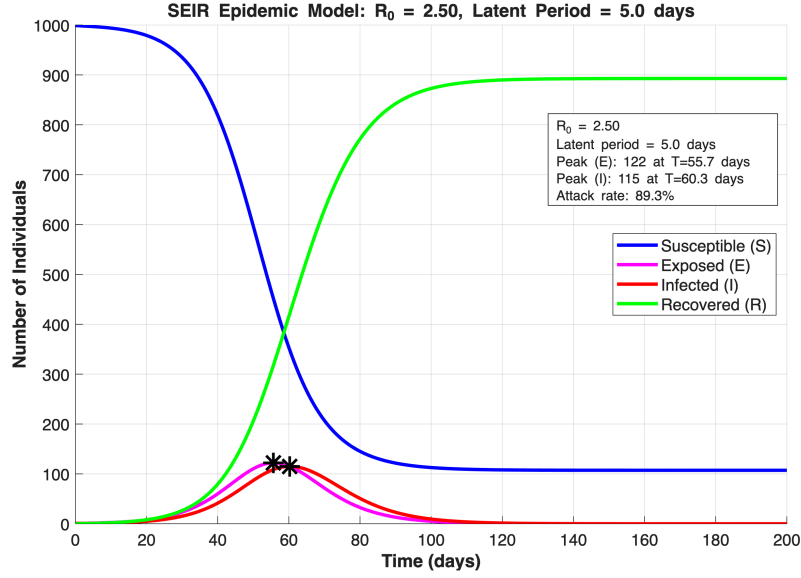


Figure 5: SEIR model showing the dynamics of susceptible (S), exposed (E), infected (I), and recovered (R) populations. The exposed compartment peaks at $T = 55.7$ days with 122 individuals, followed by the infected peak at $T = 60.3$ days with 115 individuals. The latent period of 5 days introduces a delay between exposure and infectiousness.

Figure 5 models the SEIR epidemic with $R_0 = 2.50 > 1$ and a latent period of 5.0 days, tracking the susceptible, exposed, infected, and recovered populations over a period of 200 days. The addition of the exposed compartment is the key difference between the SEIR and SIR models, as it includes a compartment for individuals who have contracted the disease but are not yet infectious E , reflecting the incubation period observed in many real world diseases. The exposed population peaks first at 122 individuals, followed shortly after by the infected population peaking at 115 individuals, shown by both asterisks. This delay between the two peaks is due to the latent period, as individuals must enter the exposed compartment before becoming infectious. The susceptible population declines throughout, while the recovered population grows and stabilises.

Compared to the SIR model, the latent period delays and slightly reduces the epidemic peak, as the exposed compartment slows the rate at which new infectious individuals enter the population.

2.4.4 A Modified SEIR Model for COVID-19

While the basic SEIR model provides a foundation for studying epidemic dynamics, its limitations mean that adaptations are often necessary to capture real world disease behaviour. For this reason, many researchers have adapted this framework to better capture COVID-19 dynamics. He et al. [23], for example, proposed a modified SEIR model that extends the classic structure by incorporating two different infectious groups, a quarantine class, and a hospitalisation class into the SEIR model to better capture COVID-19's dynamics.

The adaptations of the proposed model are the subdivision of the infectious compartment into two classes, which gives a clear distinction between those who are infectious without intervention (I_1) and those infectious with intervention (I_2), alongside these compartments, we also have the quarantined (Q) and hospitalised (H) compartments. This allows the model to capture the impact of public health measures such as isolation, which played a central role in managing COVID-19 transmission in the United Kingdom [32]. The model also incorporates waning immunity modelled by parameter ω , during the pandemic, loss of immunity was observed in many individuals[48]. Therefore, including this in the model allows it to better capture COVID-19 dynamics.

Building on this framework, the model proposed in this dissertation extends the He et al. structure by incorporating additional compartments; a vaccinated class V and a deceased class D . The inclusion of a vaccinated compartment is due to the rollout of COVID-19 vaccines beginning in December 2020 [15], which introduced a new route where susceptible individuals could gain immunity from the infection. The deceased compartment is included due to the significant number of deaths caused by COVID-19 [39]. Tracking deaths through a deceased class D allows the model to distinguish between recovery and death as different outcomes of infection. The model also keeps the waning immunity mechanism from the model by He et al., where recovered individuals may return to the susceptible class at rate ω [23], thus reflecting the observed loss of immunity over time during the COVID-19 pandemic [48].

The system variables for the proposed model can be summarised as

Variable	Description
S	Susceptible
E	Exposed class
I_1	Infectious without intervention
I_2	Infectious with intervention
R	Recovered
Q	Quarantined
H	Hospitalised
V	Vaccinated
D	Deceased

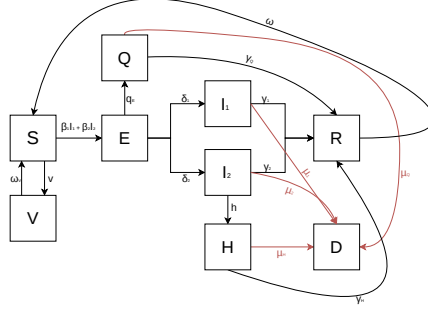
Table 1: System variables

Where $N = S + E + I_1 + I_2 + R + Q + H + V + D$ denotes the total population. The inclusion of D means that N is a constant throughout the model; this implies that individuals who die move to the compartment D rather than being removed from the population, therefore keeping N constant.

The full system of ordinary differential equations governing the extended model is given by:

$$\begin{aligned}
\frac{dS}{d\tau} &= -\beta_1 S I_1 - \beta_2 S I_2 - vS + \omega R + \omega_V V \\
\frac{dE}{d\tau} &= \beta_1 S I_1 + \beta_2 S I_2 - (\delta_1 + \delta_2 + q_E) E \\
\frac{dI_1}{d\tau} &= \delta_1 E - (\gamma_1 + \mu_1) I_1 \\
\frac{dI_2}{d\tau} &= \delta_2 E - (\gamma_2 + \mu_2 + h) I_2 \\
\frac{dQ}{d\tau} &= q_E - (\gamma_Q + \mu_Q) Q \\
\frac{dH}{d\tau} &= h I_2 - (\gamma_H + \mu_H) H \\
\frac{dR}{d\tau} &= \gamma_1 I_1 + \gamma_2 I_2 + \gamma_Q Q + \gamma_H H - \omega R \\
\frac{dV}{d\tau} &= vS - \omega_V V \\
\frac{dD}{d\tau} &= \mu_1 I_1 + \mu_2 I_2 + \mu_Q Q + \mu_H H
\end{aligned} \tag{1}$$

We can represent the movement between compartments as:



The system of equations has the following parameters

Parameters	Description
β_1	The transmission rate from the infected class without intervention
β_2	The transmission rate from the infected class with intervention
δ_1	Represents the movement from exposed to infectious without intervention
δ_2	Represents the movement from exposed to infectious with intervention
q_E	Rate of movement for exposed to quarantine
v	The rate at which an individual moves from susceptible to vaccinated
ω	The rate a recovered individual can be reintroduced to the susceptible population
ω_V	The rate at which a vaccinated individual can be reintroduced to the susceptible population
h	Hospitalisation rate from I_2 to H
γ_1	Represents the recovery rate from I_1
μ_1	Represents the rate of death from I_1
γ_2	The rate of recovery from I_2
μ_2	Death rate from I_2
γ_Q	The rate of recovery from Q
μ_Q	Death rate from Q
γ_H	The rate of recovery from H
μ_H	Death rate from H

Table 2: System Parameters

2.4.5 Extended SEIR Simulation

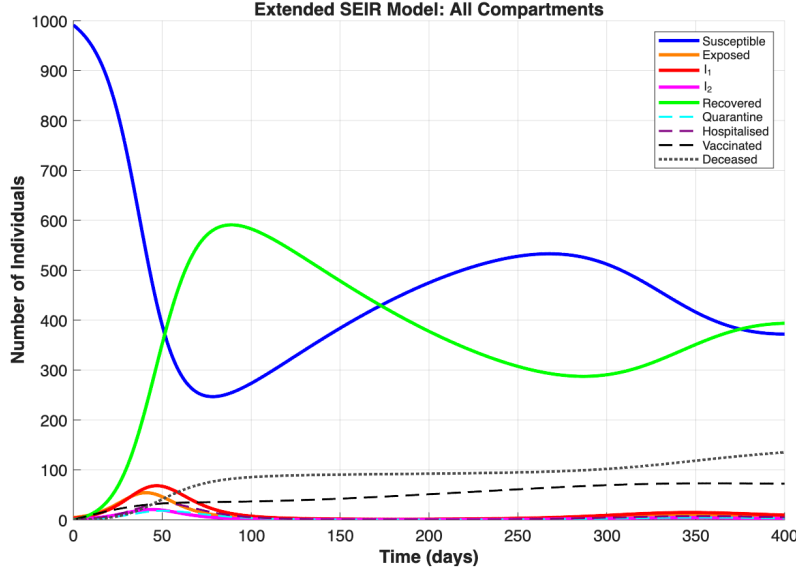


Figure 6: Extended SEIR model showing the dynamics of susceptible (S), exposed (E), infected (I), recovered (R), quarantine (Q), hospitalised (H), vaccinated (V), and deceased (D) populations. Over 365 days for a population of 1000 individuals.

Figure 6 is the extended SEIR model for a population of 1000 individuals over a 400 day period. Individuals in I_1 transmit at a rate of $\beta_1 = 0.5$, while individuals with intervention I_2 transmit at a reduced rate of $\beta_2 = 0.15$, showing the effect of public health interventions on transmission rates.

The first wave of the epidemic resolved at approximately 75 days, the exposed and infectious compartments peak early at approximately 50 days, with the susceptible population declining sharply as individuals move through E into I_1 and I_2 . These dynamics are similar to the SEIR model observed earlier. However, the recovered compartment R rises rapidly during the first wave, before steadily declining. This decline in R is because of the waning immunity term ω in the model, which returns recovered individuals, who have survived, back to the susceptible compartment at rate $\omega = \frac{1}{182}$, the value represents half a year of immunity. The vaccinated compartment V and deceased compartment D grows steadily throughout. We see a sharper increase in the deceased compartment compared to the vaccinated compartment. This is because susceptible individuals are vaccinated at rate $v = 0.001$, but experience waning at rate $\omega_V = \frac{1}{182}$ which means they are reintroduced back into the susceptible population.

2.5 Limitations of Traditional Models

Compartmental models are helpful in understanding epidemic dynamics and informing public health responses. However, the COVID-19 pandemic exposed the limitations of these mathematical approaches [34], understanding these limitations is important. This is because we can better understand model predictions and recognise when complementary methods, such as machine learning approaches, need to be used.

These models simplify complex disease dynamics. An assumption of all compartmental models is homogeneous mixing, where every individual in the population has an equal probability of coming into contact with any other individual. In reality, contact patterns vary, shaped by factors such as age and location [13], this was seen during COVID-19, as transmission were heavily concentrated in healthcare settings and care homes [10]. Therefore by treating the population as a single, uniformly mixing group, compartmental models underestimate outbreaks.

Compartmental models also treat transmission and recovery rates as fixed constants throughout an epidemic [8]. This assumption does not work in practice as many infections evolve new variants which cause parameters to shift over time. Models that cannot adapt to changes dynamically will produce predictions that differ from observed data as the epidemic progresses.

The predictive accuracy of a compartmental model is reliant on how dependable the data of an infection’s initial conditions and parameters are. This was evident during the early stages of COVID-19, as there was significant under reporting of cases and delays in death reporting, which therefore introduced doubt into parameter estimation [43].

Finally, traditional models group individuals into broad compartments, ignoring the individual differences that influence disease outcomes. During COVID-19, factors such as age and pre existing health conditions influenced the clinical outcomes within the infected compartment [16]. For example, a healthy young person and an elderly patient with underlying health conditions would be placed in the same infected compartment, despite having different risks.

3 Machine Learning Approaches to Epidemic Modelling

3.1 Introduction

Machine learning complements compartmental modelling of infectious diseases. While compartmental modelling provides an insight into disease dynamics, there are some limitations when applied to real world epidemic data [49]. Machine learning methods can remove some of the limitations that compartmental modelling faces by predicting the evolution and spread of pathogens both spatially and temporally [47].

The main advantages of machine learning in epidemic modelling include the way it handles large amounts of data and captures changes in disease dynamics [12]. When handling real world data, parameters such as transmission rates β and recovery rates γ vary, this is because of factors including changes in human behaviour, implementation of public health interventions, and the infection evolving [30]. Machine learning models can identify these patterns in epidemiological data without requiring researchers to define every biological or social mechanism influencing disease spread.

During the COVID-19 pandemic, public health agencies collected large amounts of data, including: daily case counts, hospitalisation rates, testing volumes, social media sentiment, and vaccination coverage [50]. Putting all of these data sources into a traditional compartmental framework is challenging. As it requires specifying how each factor influences transmission through various ordinary differential equations (ODEs). Machine learning models are able to learn which features are most predictive of future epidemic trends and how these features interact to influence disease outcomes [12].

3.2 Machine Learning Fundamentals for Epidemic Prediction

To understand how machine learning can improve epidemic modelling and prediction, we must first introduce linear regression and logistic regression as approaches to epidemic prediction.

3.2.1 Linear Regression

Linear regression is the foundational parametric model for supervised learning [31]. In terms of epidemic modelling linear regression provides a method to estimate how epidemic outcomes depend on given data.

The training dataset can be given by $\Gamma = (x_i, y_i)_{i=1}^n$ [31] where x_i might represent a vector of features such as COVID tests conducted. The model assumes the output can be described as a linear combination of input variables plus a

noise term ϵ , which can also be called the error term. The linear regression model can be represented mathematically by:

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_p x_p + \epsilon = \theta^T x + \epsilon$$

where $\theta = [\theta_0, \theta_1, \dots, \theta_p]^T$ is the parameter vector to be learnt from data [31].

To estimate the parameter vector θ from the training data, we must define a loss function, $L(\hat{y}, y)$, that measures the difference between the model's prediction $\hat{y}(x; \theta) = \theta^T x$ and the observed output y . For epidemics, this can represent the difference between a model's predicted daily case count and the reported figure. The best choice to calculate this difference is the squared error loss function:

$$L(\hat{y}(x; \theta), y) = (\hat{y}(x; \theta) - y)^2$$

Averaging the squared error loss function over all n training points defines the cost function $J(\theta)$, which represents the overall prediction error of the model across the entire data. To find the parameter vector θ that brings the model's count closest to the true observed figures, we find the value of θ that makes $J(\theta)$ as small as possible:

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^n (\theta^T x_i - y_i)^2 = \underset{\theta}{\operatorname{argmin}} \frac{1}{n} \|X\theta - y\|_2^2$$

where $\underset{\theta}{\operatorname{argmin}}$ is the value of θ at which the cost function is minimised. This is the least squares problem [31], and when $X^T X$ is invertible, it admits the closed form solution via the normal equation:

$$\hat{\theta} = (X^T X)^{-1} X^T y$$

This closed form solution is an advantage of linear regression over more complex models because it is easier to interpret. However, the assumption of a linear relationship between inputs and outputs is a limitation for epidemic modelling, as case trajectories are nonlinear [11]. This is the reason to look at more complex machine learning approaches, such as neural networks.

3.2.2 Logistic Regression

While linear regression predicts a continuous output, logistic regression models the probability of one of two possible outcomes [31]. In epidemiology, this might be whether a patient recovers or dies, or whether an individual becomes hospitalised given their health characteristics. This is important as it broadens the range of questions that can be addressed mathematically, moving from just predicting case counts to classifying disease outcomes at an individual level [51].

Given the set of input features $X = [x_1, x_2, \dots, x_p]^T$, where each x represents a measurable amount that can be either daily case counts or hospitalisation rates. Logistic regression models the probability that an observation belongs to the positive class as:

$$p(y = 1|x; \theta) = \sigma(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$

where σ is the logistic (sigmoid) function, which maps any real value input to a probability between 0 and 1. The parameter θ in the case of logistic regression is estimated by maximising the log likelihood over the training data. What this means is that if a logistic regression model were to be trained on COVID-19 patient data, it could estimate the probability of hospitalisation given individual characteristics.

3.3 Neural Networks for COVID-19 Prediction

Artificial neural networks (ANNs) are inspired by the signalling behaviour of neurones in biological neural networks [6]. These networks can analyse complex interactions among measurable disease related data and predict disease outcomes [52]. During the COVID-19 pandemic, neural networks were used to forecast case trajectories from large datasets that would be difficult to model using the compartmental frameworks [12].

The basic building block of a neural network is the artificial neurone, defined by the function:

$$f^{(1)}(\mathbf{x}) = a \left(w_0^{(1)} + \mathbf{w}^{(1)} \cdot \mathbf{x} \right)$$

where $\mathbf{w}^{(1)}$ is the weight vector, $w_0^{(1)}$ is a bias term, and $a : \mathbb{R} \rightarrow \mathbb{R}$ is an activation function [31]. Common activation functions include the sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$, which is related to the logistic function used for classifying disease outcomes, and the rectified linear unit $a(z) = \max(0, z)$.

A single neurone is limited in its ability to capture the nonlinear relationships present in epidemic data, such as the interaction between transmission rates and intervention policies that influenced COVID-19 spread. Deep neural networks address this by stacking multiple layers, where the output of layer l is:

$$\mathbf{h}^{(l)} = a \left(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)} \right)$$

with $\mathbf{h}^{(0)} = \mathbf{x}$ as the input data [31]. The network is trained by minimising a loss function, such as the mean squared error:

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

using backpropagation, which computes gradients via the chain rule, and gradient descent, which updates the weights iteratively as $\mathbf{W}^{(l)} \leftarrow \mathbf{W}^{(l)} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(l)}}$, where η is the learning rate [31]. A key challenge is overfitting, where the network memorises training data rather than learning the underlying disease

dynamics. This is a risk in epidemic modelling, where data from the early stages of an outbreak may be limited due to under-reporting [43]. Regularisation techniques such as dropout, which randomly removes neurones during training, help to prevent this [6].

Since epidemic case data is in order of time, recurrent neural networks (RNNs) are a natural architecture for COVID-19 forecasting, where inputs such as daily case counts are used to predict future values. The most used variants are long short term memory and gated recurrent unit networks [12]. These architectures use gating mechanisms to retain information over long sequences such as epidemic data, making them suited for modelling epidemic trajectories where case counts may depend on intervention policies.

3.4 Hybrid Models: Combining Compartmental Models with Machine Learning Tools

Hybrid models combine the strengths of both machine learning and traditional epidemiological models. By merging the structure of compartmental models with the data driven, predictive tools of machine learning, hybrid models give a more accurate representation of disease spread. These models improve the estimation of parameters and incorporate real time data, which improves decision making during disease outbreaks.

The reason for hybrid approaches is that neither compartmental models nor machine learning methods are fully sufficient on their own. Compartmental models such as SIR and SEIR assume fixed parameters, such as transmission rates, recovery rates, and contact patterns, which may not stay constant as a pandemic progresses through different phases. Purely data driven machine learning models can capture complex nonlinear patterns in case data, but lack the understanding of epidemics that makes compartmental models so valuable for understanding transmission dynamics and evaluating intervention strategies.

3.4.1 Hybrid Modelling of COVID-19

In their paper Zhao et al. (2022) [57] demonstrated hybrid modelling by combining a logistic growth model with the SEIR framework to analyse COVID-19 transmission across Pakistan. The method used parameter estimation from the logistic model to initialise the SEIR model’s compartments. This reduces the number of parameters that require manual input. Parameter adaptation in SEIR modelling is a time consuming process which requires repeated adjustment to achieve accurate predictions. Therefore, by having this done by logistic regression means that parameter estimation can be automated and updated from observed data, reducing manual tuning while improving the responsiveness of the model to changing epidemic dynamics.

This model shows how machine learning can be used alongside comparten-

tal models without replacing their structure; instead, enhancing their flexibility and predictive accuracy by integrating data driven techniques to inform key parameters while preserving the epidemiological framework.

3.4.2 Physics Informed Neural Networks

More recently, physics informed neural networks (PINNs) have emerged as a promising framework for hybrid modelling. PINNs embed the differential equations of a compartmental model directly in the loss function. This means the network is trained not only to look at observed data but also to satisfy mathematical constraints of the epidemiological system [44]. Shahmohammadi et al. [36] applied this approach to a modified SEIR model for COVID-19, using a neural network to approximate the system of ordinary differential equations and a separate network to estimate the transmission parameters. Therefore, this addresses one of the most significant challenges of compartmental modelling the assumption of constant parameters. By allowing the parameters to vary over time, the model can capture the effects of changes in interventions, public behaviour, and the emergence of new variants, all of which are difficult to represent in a standard ODE framework.

The advantage of PINNs over compartmental approaches is that the epidemiological equations serve as a regulator, guiding the neural network to produce true outputs. This means that even when data is limited, the model is less likely to generate biologically unrealistic predictions.

3.4.3 Hybrid Modelling Strategies

These studies show that hybrid models can be constructed through different strategies. These include using machine learning to estimate initial conditions for a compartmental model or fully integrating the compartmental equations into the learning process itself through PINNs. Each approach addresses different limitations of machine learning tools and compartmental models, such as the inability to capture complex disease dynamics.

4 Modelling COVID-19 Using Mathematical Approaches

4.1 COVID-19: Epidemiological Context and Data

The United Kingdom experienced two distinct waves of SARS-CoV-2 transmission during 2020 and 2021[38]. The first wave is defined as the period from 31 January 2020 to 6 September 2020, and the second wave from 7 September 2020 to 12 April 2021 [53], providing a documented epidemiological dataset that is ideal for compartmental modelling.

During the first wave, NHS England reported approximately 309,000 confirmed cases and over 50000 deaths[50]. Interventions, such as the national lockdown imposed on 23 March 2020 [27], played a central role in reducing transmission. The second wave was larger in scale, with confirmed cases exceeding 3,500,000 [50]. The rollout of vaccination, beginning on 8 December 2020, introduced another dynamic to the epidemic, which makes the second wave interesting from a modelling perspective.

For this study, daily case data, death counts, and recovery estimates were obtained from publicly available datasets maintained by the UK Government [39]. Specifically, the key variables extracted are: daily confirmed cases $C(\tau)$, cumulative deaths $D(\tau)$, and estimated recoveries $R(\tau)$, where τ represents time in days from the start of the observed period. The total UK population is taken as approximately $N = 67100000$ based on 2020 estimates from the Office for National Statistics [40].

For the following analysis, the focus is on the first wave of infections as the base scenario for the standard SEIR model, as this period predates vaccination. The second wave is then used to evaluate the extended model developed earlier, incorporating vaccination, quarantine, hospitalisation, and waning immunity.

4.2 SEIR Model Implementation for COVID-19

The standard SEIR model is applied to England’s COVID-19 data. The system of ordinary differential equations is solved numerically within MATLAB over a time $T = 219$ days, as this fits the exact start and end values of the first wave of infections. Initial conditions are derived from early case data: $I(0)$ is estimated from the first reported cases approximately 20000 and an assumed ratio of exposed to infectious individuals, $R(0) = 0$, and $S(0) = N - E(0) - I(0)$, with $N = 67100000$.

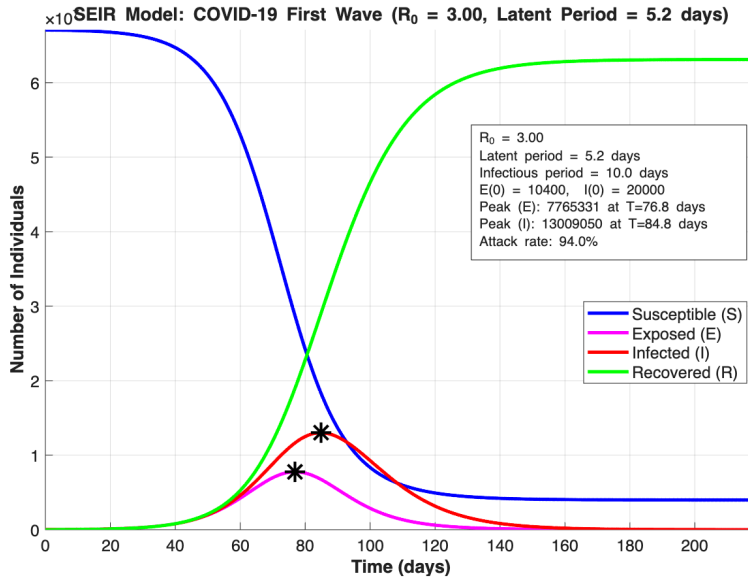


Figure 7: SEIR model fitted to the first wave of COVID-19 in England with $R_0 = 3.00$ and a latent period of 5.2 days. The model captures the timing of the epidemic peak at approximately 85 days from the start of the wave. The susceptible population declines sharply as individuals move through the exposed and infected compartments, with the recovered population stabilising by the end of the simulation period.

Figure 7 shows the first wave of COVID-19 fitted to the basic SEIR model, capturing the timing of the epidemic peak at approximately April 2020, calculated from when the first wave began. The estimated basic reproduction number $R_0 = \frac{\beta}{\gamma}$ falls within the range of 2.5 to 3.5 reported in the early COVID-19 [3], and the mean incubation period $\frac{1}{\delta}$ is consistent with the 5 to 6 day period observed during the pandemic [42]. However, the standard SEIR model underestimates the epidemic curve, as mechanisms such as quarantine, hospitalisation,

and vaccination, are not included in this model. This is why the use of the extended model in the following sections captures COVID-19 disease dynamics better.

4.3 Intervention Scenarios

To assess the impact of public health interventions on the second waves epidemic curves, the extended $S-E-I_1-I_2-Q-H-R-V-D$ model introduced in Section 2.4.4 is used. This is because this wave differs from the first wave, due to the inclusion of the COVID-19 vaccination rollout. Three different scenarios are simulated to show the contribution of specific control measures.

The initial scenario applies no vaccination and no quarantine, representing the undisturbed spread of COVID-19 with transmission parameters fitted to the second wave data. The second scenario examines the effects of vaccination on second wave data, by introducing vaccination at a rate v . Waning immunity is modelled as a transition from compartment V back to S at a rate $\omega = \frac{1}{183}$, which represents approximately six months of protection provided by the vaccination [55]. The final scenario additionally activates the quarantine pathway at a rate δ_q , reflecting the effectiveness of the COVID-19 test and trace programmes, alongside hospitalisation from severe infections.

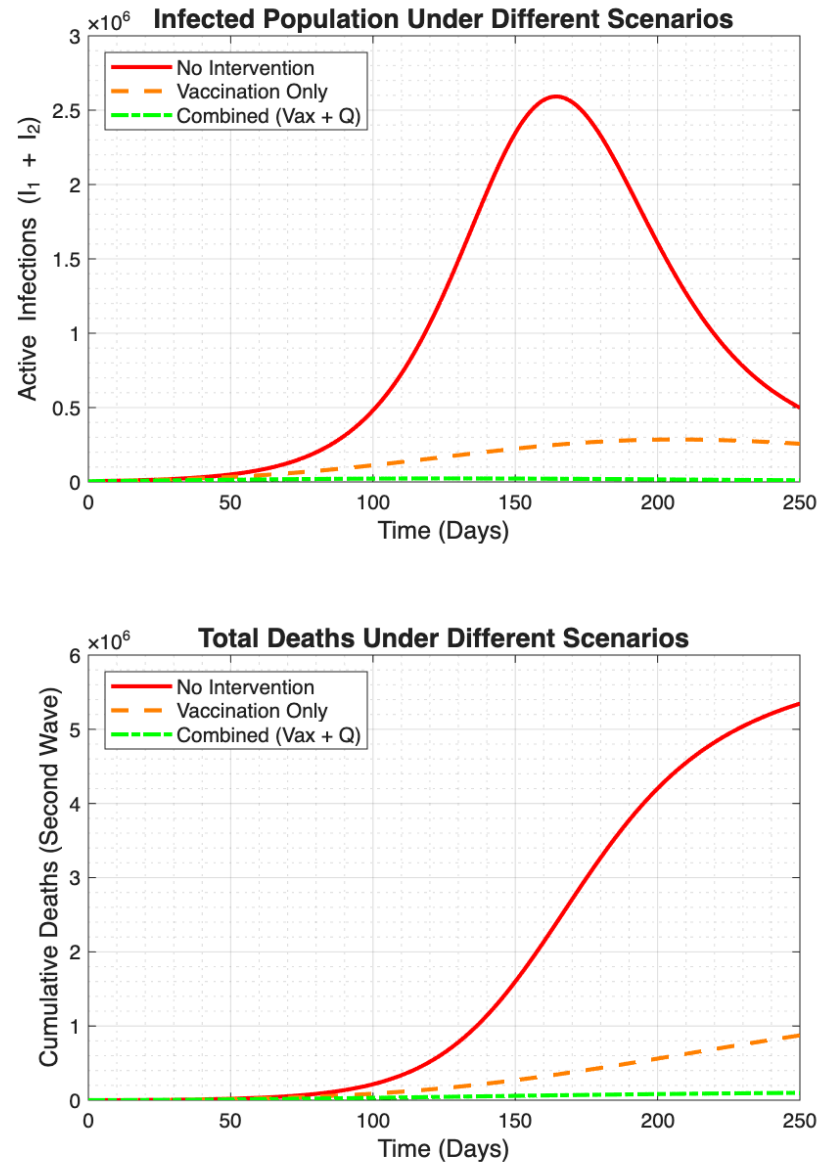


Figure 8: The effects of intervention strategies on the spread and total deaths of COVID-19

Figure 8 illustrates that vaccination reduces both peak infections and total mortality compared to no interventions. However, the combination of quarantine and vaccination gives an even greater reduction of the epidemic curve. Therefore, when vaccination, quarantine, and recovery act together, R_0 is driven below one faster than any single measure can achieve on its own.

4.4 Model Validation and Sensitivity Analysis

To evaluate the extended model, a sensitivity analysis is performed on the model's parameters. Each parameter is varied within $\pm 10\%$ of its value while all others are held constant, and the effect on total infections over the period is calculated.

The parameters examined are the transmission rate from I_1 (β_1), the rate of progression from exposed to infectious without intervention (δ_1), the recovery rate from I_1 (γ_1), the vaccination rate (v), rate from quarantine to exposed (q_E), and the hospitalisation rate (h).

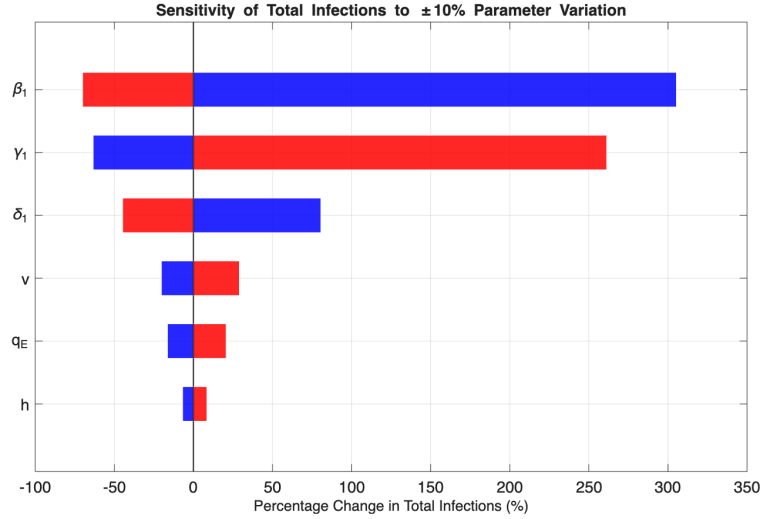


Figure 9: Sensitivity analysis of key parameters in the Extended Model

Figure 9 demonstrates that the model is most sensitive to β_1 , which produces the largest swing in total infections under varied parameters. This is expected since β_1 governs the transmission pathway from individuals infected without intervention, who have unrestricted contact with the susceptible population. The recovery rate γ_1 is the second most influential parameter, as it directly governs the duration of the infectious period for I_1 . The vaccination rate v is third,

confirming that even moderate changes in rollout speed have an impact on epidemics.

The quarantine rate q_E and the hospitalisation rate h have marginal effect on total infections. The progression rate δ_1 has a moderate effect on total infections. In particular, reducing δ_1 delays the epidemic peak and reduces cumulative infections. This reflects the role of the exposed compartment as a buffer that slows the rate at which new infections enter the transmission chain.

5 Eradication and Control Strategies

5.1 Introduction

The compartmental models describe how infectious diseases spread through populations. From a public health perspective, the goal is to control or eradicate the infections [19]. This chapter examines the mathematical conditions under which disease eradication is achievable and analyses the effectiveness of key control strategies, including vaccination and interventions. By connecting these strategies to the basic reproduction number R_0 , we establish quantitative criteria that public health authorities can use to guide decision making during an epidemic with a focus on COVID-19.

While compartmental based analysis from the earlier chapters provides the theoretical conditions for disease control, it does not address the practical question of how to assign resources to achieve this. In real epidemics, policymakers face decisions about when to impose lockdowns, how quickly to roll out vaccines across different population groups, and when interventions can safely be relaxed [25]. These problems are often difficult to solve using the compartmental framework alone. Machine learning, in particular neural network based optimisation and hybrid models, offers a data driven approach to these questions by learning control policies directly from simulated or observed epidemic data [12].

Throughout this chapter, we build directly on the SEIR framework from Chapter 2, the COVID-19 specific model from Chapter 4, and some machine learning foundations established in Chapter 3.

5.2 Vaccination Strategies and Herd Immunity

Vaccination is one of the most effective tools for controlling infectious diseases [20]. From a mathematical perspective, vaccination works by transferring individuals from the susceptible compartment S directly to the recovered (immune) compartment R , thus reducing the number of susceptible individuals available for infection. In the extended model developed in Section 2.4.4, this movement is modelled by the vaccination rate v , which moves individuals from S to the vaccinated compartment V .

Vaccination can be expressed as a fraction where v represents the proportion of the population that has been vaccinated; with a vaccine giving 100% immunity, the new reproduction number under vaccination becomes

$$R_v = (1-v)R_0.$$

[9] this expression shows that as vaccination coverage v increases, the reproduction number R_v decreases.

The definition of herd immunity originates from mathematical frameworks for the impact of vaccination on models [9]. Setting $R_v = 1$ and solving gives the herd immunity level h_C which can be defined by the following:

$$h_C = 1 - \frac{1}{R_0}$$

where R_0 is the basic reproduction number. During the COVID-19 pandemic, early predictions of R_0 were between 2 and 3 [3]. This means that the herd immunity range is $0.50 \leq h_C \leq 0.67$, which meant that between 50% and 67% of the population needed to be immune to prevent transmission.

The herd immunity threshold shown above assumes that immunity once gained is permanent. However, the extended SEIR model in this dissertation includes a waning immunity term ω that returns recovered individuals to the susceptible compartment S at rate ω which makes the model more accurate for COVID-19. This means the immune fraction h_c is not a fixed total. If immunity declines faster than new individuals are vaccinated, the susceptible population grows; R_v goes back above 1, leading to further waves of infection. This is consistent with the pattern observed in the UK [21], where multiple waves occurred despite increasing vaccination coverage. This was driven by waning immunity and the emergence of new variants [48]. Moore et al. [35] demonstrated this using a model fitted to UK epidemiological data. Their analysis showed that even under the most ideal conditions, with a vaccine preventing 85% of reinfections and a high proportion of the population vaccinated, the reproduction number after full removal of interventions remained at $R_0 = 1.58$. This showed that achieving $R_v < 1$ at a point in time is not sufficient for long term control.

5.2.1 Simulation of Vaccines

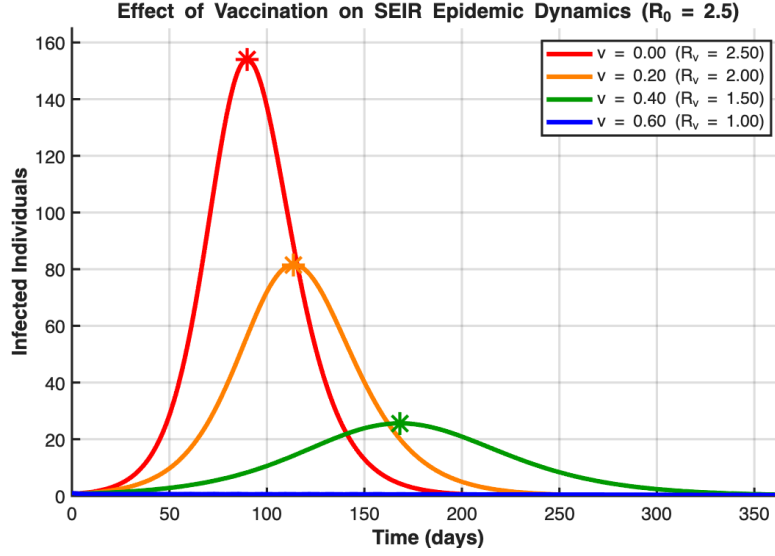


Figure 10: Model showing the effect of vaccination on SEIR epidemics with $R_0 = 2.50$

Figure 10 shows the effect of vaccination on SEIR epidemic curves over a period of $T = 365$ days. With no vaccination ($v = 0.00$), the number of infected individuals increases rapidly to a high peak before declining; this produces a standard epidemic curve. As the vaccination rate increases ($v = 0.20$ and $v = 0.40$), the peak number of infections is reduced and occurs later in time shown by the asterisk at the top of the curve. For a higher vaccination rate ($v = 0.60$), the number of infected individuals remains close to zero throughout the simulation, indicating that the epidemic is effectively suppressed. This simulation shows that increasing vaccination reduces transmission and can prevent large outbreaks, in SEIR epidemics.

5.3 Interventions

Interventions such as lockdowns, social distancing, and self-isolation are good examples of non pharmaceutical intervention strategies (NPIs) [18]. These strategies were used by public health agencies worldwide to reduce COVID-19 transmission rates [18]. Mathematically, these strategies work by directly reducing the rate of contact between susceptible and infectious individuals, thereby lowering the transmission rate β [17]. During the COVID-19 pandemic, the United Kingdom's government implemented a range of NPIs [4].

The effect of NPIs on transmission can be modelled by introducing a reduction factor, which effectively reduces β from β_n , the contact rate during normal times, to β_i , the contact rate during intervention strategies, with $\beta_i < \beta_n$ [46]. This reduction can be expressed as a piecewise function of time:

$$\beta(\tau) = \begin{cases} \beta_n, & 0 \leq \tau < \tau_s, \\ \beta_i, & \tau_s \leq \tau < \tau_s + \tau_d, \\ \beta_n, & \tau_s + \tau_d \leq \tau \end{cases}$$

where τ_s is the time the intervention starts and τ_d is the duration of the intervention [46]. Once the intervention is lifted at time $\tau_s + \tau_d$, the transmission rate returns to β_n .

Since the basic reproduction number R_0 is proportional to β , NPIs reduce the reproduction number by the same factor. Under an intervention period, the reproduction number becomes $\frac{\beta_i}{\beta_n} R_0$ [46].

From this expression, we can observe that NPIs work by only reducing transmission while other measures, such as vaccination remain in effect. Once interventions are lifted, β returns to β_n and the reproduction number will increase if the infection is still present in the population. This distinction between vaccination and NPIs is key for long term epidemic planning.

In reality, the contact rate during intervention periods β_i is not fixed throughout the entire epidemic. During the COVID-19 pandemic, the United Kingdom imposed and eased lockdown restrictions in phases during 2020 and 2021 [27]. This meant that the reduced rate of transmission varied throughout the pandemic.

5.4 Data Driven Approaches to Epidemic Control

The piecewise transmission function for reduction assumes that interventions begin and end at known times with fixed effect sizes. In reality, the timing, duration, and intensity of control measures are decisions that must be made under uncertainty using incomplete and changing data [22]. This is where the machine learning methods offer an approach that works with compartmental models.

Neural network-based models can be trained on simulated or observed epidemics to learn control policies that minimise government objectives, such as total infections [12]. Rather than assuming a fixed β_i during intervention periods, neural networks can adapt the level of restriction dynamically in response to changing data.

Physics informed neural networks are relevant in understanding epidemic control. This is due to the embedding of the ODE structure of a compartmental model within the network's loss function. PINNs can estimate time varying pa-

rameters such as $\beta(\tau)$ directly from observed data without requiring the modeller to specify when interventions began or ended [54]. This makes them well suited for retrospective analysis of epidemics where the effect of overlapping policies is difficult to separate using traditional parameter fitting alone.

These approaches do not replace the analytical results derived earlier. The herd immunity threshold h_C and the condition $R_v < 1$ remain the theoretical targets that any control strategy must achieve. Machine learning instead provides a practical framework for deciding how and when to reach those targets, given real world constraints and data.

6 Conclusion

6.1 Discussion

This dissertation has developed, analysed, and computationally simulated mathematical models for studying the spread of infectious disease, with a focus on COVID-19, beginning with the SIS and SIR models, then moving to the SEIR model, and extending the model to more accurately capture COVID-19 dynamics. The model incorporated quarantine, hospitalisation, vaccination, waning immunity, and disease induced mortality. Each model provided new epidemiological factors, allowing for a more accurate representation of the mechanisms that govern COVID-19 transmission in the United Kingdom.

Compartmental models have their limitations when looking at epidemics, but machine learning methods can complement these compartmental approaches. Linear and logistic regression were the foundation for understanding how predictive tools work. Neural networks can capture complex relationships in epidemic data. Furthermore, hybrid models embed compartmental models' ODE structure within their frameworks in order to relax the assumption of fixed parameters in conventional models.

These models were then applied to COVID-19 data, fitting the standard SEIR model to the first wave data and using the extended model to simulate intervention scenarios during the second wave. Findings showed that vaccination alone reduced peak infections and cumulative deaths, but the combination of vaccination with quarantine achieved a greater reduction in the epidemic curve. Sensitivity analysis showed that the transmission rate β_1 is the most important parameter, followed by the recovery rate γ_1 .

Finally, chapter 5 connected these modelling results to public health strategies by finding the herd immunity threshold $h_C = 1 - \frac{1}{R_0}$ and analysing the conditions under which the effective reproduction number under vaccination R_v . The inclusion of waning immunity showed that achieving $R_v < 1$ at a single point in time is insufficient for long term control of COVID-19, shown with the multiple waves observed in the UK despite rising vaccination coverage. Non pharmaceutical interventions were also explored using a piecewise reduction in β , demonstrating their role as a complementary approach with vaccination.

6.2 Key Findings

- The basic reproduction number R_0 controls epidemic threshold behaviour across compartmental models. When $R_0 > 1$ the disease spreads through the population, and when $R_0 < 1$ the infection dies out.
- The extended $S-E-I_1-I_2-Q-H-R-V-D$ model captures more COVID-19 related dynamics, including the distinct roles of intervention and non intervention infectious compartments, the effect of quarantine, and the loss of immunity over time.
- Combined interventions are more effective than single measures alone. The figures in chapter 4 showed that vaccination paired with quarantine achieved a larger decrease in peak infections and total deaths than just vaccination.
- The sensitivity analysis identified the transmission rate β_1 as the parameter with the biggest influence on total infections. This supports the government's interventions such as nationwide lockdowns to reduce contact during epidemic peaks.
- The herd immunity threshold $h_C = 1 - \frac{1}{R_0}$ gives a target for vaccination coverage. The observed waning of immunity during the COVID-19 pandemic demonstrated, that maintaining $R_V < 1$ over time requires sustained effort beyond initial vaccination.
- Machine learning approaches, particularly physics informed neural networks, allow us to address the assumption of constant parameters, by allowing transmission and recovery rates to vary in response to changing conditions.

7 Final Reflections

My original project plan was to develop compartmental models, apply them to real COVID-19 data, and implement a neural network to predict case trajectories, the plan changed significantly. While the compartmental modelling was more than what I originally wanted with the extended SEIR model, the neural network implementation did not go as planned. I made the decision to present the machine learning material as a theoretical chapter and to focus computational work on the extended SEIR model. This was the right decision for me, as it allowed me to focus more on precise compartmental modelling, the machine learning implementation proved more complex than anticipated.

One of my earliest mathematical mistakes was in deriving the basic reproduction number R_0 . When making the SIS model. It produced an expression that did not simplify to the expected ratio $\frac{\beta}{\gamma}$, which I was trying to find. It was only after revisiting Britton [8], which I followed closely throughout this project,

and reworking the algebra step by step, that I found what I was doing wrong. Revisiting Britton helped me correct my algebraic errors when I encountered difficulties with concepts.

Another challenge was when formulating the ODEs for the extended SEIR model. With nine compartments and lots of different parameters, writing the equations correctly required careful attention to where each flow enters and leaves. The most effective approach was creating diagrams so I could see the transitions between each of the compartments. Also, expanding on other people's work helped, as it provided a reference framework.

Fitting models to real COVID-19 data was harder than I thought, as government data needed to be extrapolated from publicly available datasets. What I found was that real epidemic data is underreported relative to the true figures actually are, and small changes in assumptions about initial conditions produced noticeably different epidemic curves, which meant in some simulations I had to make informed estimations on initial data. Getting MATLAB simulations to run the way I wanted them to was frustrating at times, particularly in systems where the solver produced figures and values that couldn't be correct. Learning to check my outputs against expected behaviour was a skill that I had to develop throughout the project.

This project has developed my ability to read mathematical literature to find citations, formulate and solve systems of differential equations, and evaluate the assumptions of mathematical models. The mistakes I made, from algebraic errors in deriving R_0 to the errors in formulating my ODEs, helped me learn how to overcome these challenges myself. Each challenge made me look back at the mathematics more carefully and deepen my understanding of the models I was building. The experience has given me an appreciation for the actual application of complex models to real world data and also how complex machine learning models truly are. I believe that this experience overall has not only made me a better mathematician but has also improved my presentation skills and my writing clarity.

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A MATLAB Code

A.1 R_0 Simulations

A.1.1 $R_0 < 1$

```
clear; close all; clc;

%% Model Parameters
N = 1000;
gamma = 0.2;
beta = 0.15;
R0 = beta/gamma;

% Initial conditions
I0 = 100;
S0 = N - I0;

fprintf('SIS-Model-Parameters-(R0<1):\n');
fprintf('=====n');
fprintf('Beta-( ): -%.2f-per-day\n', beta);
fprintf('Gamma-( ): -%.2f-per-day\n', gamma);
fprintf('R0:-%.2f\n', R0);
fprintf('Initial-infected:-%d-(%.1f%%)\n', I0, 100*I0/N);
fprintf('=====n\n');

%% Solve the ODE
t_span = [0 200];
t_eval = linspace(0, 200, 1000);

[t, y] = ode45(@(t,y) SIS-equations(t, y, beta, gamma, N), t_eval, [S0; I0]);

S = y(:,1);
I = y(:,2);

%% Create Figure
figure('Position', [100 100 800 600]);

% Plot infected population
plot(t, I, 'b-', 'LineWidth', 3);
hold on;

% Add horizontal line at I = 0
plot([0 200], [0 0], 'k-', 'LineWidth', 1.5, 'HandleVisibility', 'off');

% Format
```

```

xlabel('Time (days)', 'FontSize', 14, 'FontWeight', 'bold');
ylabel('Infected Population', 'FontSize', 14, 'FontWeight', 'bold');
title(sprintf('SIS Model: Disease Dies Out ( $R_0 = %.2f < 1$ )', R0), ...
      'FontSize', 16, 'FontWeight', 'bold');
grid on;
set(gca, 'FontSize', 12, 'LineWidth', 1.5);
xlim([0 200]);
ylim([0 10*1.1]);

% text annotation
text(100, 10*0.7, {sprintf('R_0 = %.2f < 1', R0), ...
                  'Disease cannot persist', ...
                  'I(t) = 0 as t → ∞'}, ...
     'FontSize', 12, 'FontWeight', 'bold', ...
     'BackgroundColor', 'white', ...
     'EdgeColor', 'black', ...
     'LineWidth', 1.5, ...
     'HorizontalAlignment', 'center');

hold off;

%% Save figure
print('fig-SIS-R0_less_than_1', '-dpng', '-r300');
fprintf('Figure saved as: fig-SIS-R0_less_than_1.png\n');
fprintf('Infection dies out to zero (disease-free equilibrium is stable)\n');
fprintf('\nFigure saved as: fig-SIS-epidemic-R < 1.png\n');

function dydt = SIS_equations(t, y, beta, gamma, N)

    S = y(1);
    I = y(2);

    % Force of infection
    lambda = beta * I / N;

    % Differential equations
    dS = -lambda * S + gamma * I;
    dI = lambda * S - gamma * I;

    dydt = [dS; dI];
end

```

A.1.2 $R_0 > 1$

```
clear; close all; clc;
```

```
%% Model Parameters
```

```
N = 1000;
```

```
gamma = 0.2;
```

```
beta = 0.6;
```

```
R0 = beta/gamma;
```

```
% Calculate endemic equilibrium
```

```
I_star = N * (1 - 1/R0);
```

```
% Initial conditions
```

```
I0 = 100;
```

```
S0 = N - I0;
```

```
fprintf('SIS-Model-Parameters-(R0>1):\n');
```

```
fprintf('=====\n');
```

```
fprintf('Beta-( ): -%.2f-per-day\n', beta);
```

```
fprintf('Gamma-( ): -%.2f-per-day\n', gamma);
```

```
fprintf('R0: -%.2f\n', R0);
```

```
fprintf('Initial-infected: -%d-(%.1f%%)\n', I0, 100*I0/N);
```

```
fprintf('Endemic-equilibrium-I*: -%.0f-(%.1f%-of-population)\n', ...
```

```
I_star, 100*I_star/N);
```

```
fprintf('=====\n\n');
```

```
%% Solve the ODE
```

```
t_span = [0 200];
```

```
t_eval = linspace(0, 200, 1000);
```

```
[t, y] = ode45(@(t,y) SIS_equations(t, y, beta, gamma, N), t_eval, [S0; I0]);
```

```
S = y(:,1);
```

```
I = y(:,2);
```

```
%% Plot
```

```
figure('Position', [100 100 800 600]);
```

```
% Plot infected population
```

```
plot(t, I, 'b-', 'LineWidth', 3);
```

```
hold on;
```

```
% Horizontal line endemic equilibrium
```

```
plot([0 200], [I_star I_star], 'r—', 'LineWidth', 2, ...
```

```

'DisplayName', sprintf('Endemic-Equilibrium'));

% Format
xlabel('Time-(days)', 'FontSize', 14, 'FontWeight', 'bold');
ylabel('Infected-Population', 'FontSize', 14, 'FontWeight', 'bold');
title(sprintf('SIS-Model: Endemic-Equilibrium ( $R_0 = %.2f > 1$ )', R0), ...
    'FontSize', 16, 'FontWeight', 'bold');
legend('Infected-I(t)', sprintf('Endemic-Equilibrium', I_star), ...
    'Location', 'southeast', 'FontSize', 11);
grid on;
set(gca, 'FontSize', 12, 'LineWidth', 1.5);
xlim([0 200]);
ylim([0 N*0.75]);

% Add text
text(100, I_star*1.3, {sprintf('R0 = %.2f > 1', R0), ...
    'Disease-persists', sprintf('I(t) = I^* = %.0f', ...
    I_star)}, 'FontSize', 12, 'FontWeight', ...
    'bold', 'BackgroundColor', 'white', 'EdgeColor', ...
    'black', 'LineWidth', 1.5, 'HorizontalAlignment', ...
    'center');

hold off;

%% Save figure
print('fig_SIS_R0_greater_than_1', '-dpng', '-r300');
fprintf('Figure-saved-as:-fig_SIS_R0_greater_than_1.png\n');

fprintf(['Infection-reaches-endemic-equilibrium-at-I* = %.0f' ...
    '-----(%1f%%-of-population)\n'], I_star, 100*I_star/N);
fprintf(' \nFigure-saved-as:-fig_SIS_epidemic_R>1.png\n');

function dydt = SIS_equations(t, y, beta, gamma, N)
    % SIS model differential equations

    S = y(1); % Susceptible
    I = y(2); % Infected

    % Force of infection
    lambda = beta * I / N;

    % Differential equations
    dS = -lambda * S + gamma * I; % Susceptibles decrease by infection
    dI = lambda * S - gamma * I; % Infected increase by infection

```

```
        dydt = [dS; dI];  
end
```

A.2 Varying R_0 on SIR Epidemic curves

```
clear; close all; clc;

N = 1000;
gamma = 0.2;
I0 = 1;
R0_init = 0;
S0 = N - I0;

t_span = [0 200];
t_eval = linspace(100, 200, 1000);

%% Different R0 values
R0_values = [0.8, 1.5, 2.5, 4.0];

colors = [0 0 1;
          1 0 0;
          0 1 0;
          0 0 0];

fprintf('Comparing SIR Dynamics for Different R0 Values\n');
fprintf('=====n\n');

% Store results
results = struct();

figure('Position', [100 100 800 600]);
hold on;

for i = 1:length(R0_values)
    R0_val = R0_values(i);
    beta = R0_val * gamma;

    y0 = [S0; I0; R0_init];
    [t, y] = ode45(@(t,y) ode_SIR(t, y, beta, gamma, N), t_eval, y0);

    I = y(:,2);
    [peak, peak_idx] = max(I);
    peak_time = t(peak_idx);

    % Store results
    results(i).R0 = R0_val;

    results(i).peak = peak;
```

```

results(i).peak_time = peak_time;

results(i).final_size = y(end,3);

results(i).t = t;

results(i).I = I;

%% Plot
if R0_val < 1
    plot(t, I, '—', 'LineWidth', 2.5, 'Color', colors(i,:), ...
         'DisplayName', sprintf('R_0 = %.1f (no epidemic)', R0_val));
else
    plot(t, I, '—', 'LineWidth', 2.5, 'Color', colors(i,:), ...
         'DisplayName', sprintf('R_0 = %.1f', R0_val));
end
end

xlabel('Time (days)', 'FontSize', 14, 'FontWeight', 'bold');
ylabel('Infected Individuals', 'FontSize', 14, 'FontWeight', 'bold');
title('Effect of R_0 on Epidemic Peak', 'FontSize', 16, 'FontWeight', 'bold');
legend('Location', 'northeast', 'FontSize', 10);
grid on;
set(gca, 'FontSize', 11);
hold off;

fprintf('\nFigure saved as: fig_SIR_endemic_R0_varying.png\n');

```

A.3 SEIR Model Simulation

```

clear; close all; clc;

% Population
N = 1000;

beta = 0.5;           % Transmission rate (per day).
sigma = 1/5;          % Rate of progression from E to I.
gamma = 0.2;          % Recovery rate.

% Calculate R0
R0 = beta/gamma;

% Average time in exposed state
latent_period = 1/sigma;
% Average time in infected state
infectious_period = 1/gamma;

I0 = 1;               % Initial infected
E0 = 0;               % Initial exposed (none at start)
R0_init = 0;          % Initial recovered
S0 = N - E0 - I0 - R0_init; % Initial susceptible

% Time parameters
t_span = [0 200];
t_eval = linspace(0, 200, 1000);

fprintf('SEIR-Model-Parameters:\n');
fprintf('=====\n');
fprintf('Population-(N):-%d\n', N);
fprintf('Transmission-rate-( ):-%.3f-per-day\n', beta);
fprintf('Latent-period:-%.1f-days\n', latent_period);
fprintf('Infectious-period:-%.1f-days\n', infectious_period);
fprintf('Progression-rate-( ):-%.3f-per-day\n', sigma);
fprintf('Recovery-rate-( ):-%.2f-per-day\n', gamma);
fprintf('Basic-reproduction-number-( R ):-%.2f\n', R0);
fprintf('Initial-conditions:-S=%d,-E=%d,-I=%d,-R=%d\n', ...
        S0, E0, I0, R0_init);
fprintf('=====\n\n');

```

```

%% Solve the ODE System
y0 = [S0; E0; I0; R0_init];

% Use ODE45 solver
[t, y] = ode45(@(t,y) ode_SEIR(t, y, beta, sigma, gamma, ...
    N), t_eval, y0);

% Compartments
S = y(:,1); % Susceptible
E = y(:,2); % Exposed
I = y(:,3); % Infected
R = y(:,4); % Recovered

%% Find key metrics
[peak_exposed, peak_E_idx] = max(E);
peak_E_time = t(peak_E_idx);

[peak_infected, peak_I_idx] = max(I);
peak_I_time = t(peak_I_idx);

final_recovered = R(end);
attack_rate = (final_recovered / N) * 100;

fprintf('Epidemic-Outcomes:\n');
fprintf('===== \n');
fprintf(['Peak-exposed: -%.0f individuals - (%.1f%% of -' ...
    'population)\n'], peak_exposed, (peak_exposed/N)*100);
fprintf('Time-to-peak-exposed: -%.1f-days\n', peak_E_time);
fprintf(['Peak-infected: -%.0f individuals - (%.1f%% of -' ...
    '----population)\n'], peak_infected, (peak_infected/N) ...
    *100);
fprintf('Time-to-peak-infected: -%.1f-days\n', peak_I_time);
fprintf('Time-delay-(E-peak-to-I-peak): -%.1f-days\n', ...
    peak_I_time - peak_E_time);
fprintf('Final-attack-rate: -%.1f%%\n', attack_rate);
fprintf('Duration-of-epidemic: -~%.0f-days\n', ...
    t(find(I > 1, 1, 'last')));
fprintf('===== \n');

%% Create figure
figure('Position', [100 100 900 600]);

% Main plot: all four compartments
hold on;
plot(t, S, 'b-', 'LineWidth', 2.5, 'DisplayName', ...
    'Susceptible-(S)');

```

```

plot(t, E, 'm-', 'LineWidth', 2.5, 'DisplayName', ...
    'Exposed-(E)');
plot(t, I, 'r-', 'LineWidth', 2.5, 'DisplayName', ...
    'Infected-(I)');
plot(t, R, 'g-', 'LineWidth', 2.5, 'DisplayName', ...
    'Recovered-(R)');

% Mark peak exposed and infected
plot(peak_E_time, peak_exposed, 'black*', ...
    'MarkerSize', 15, 'LineWidth', 2, ...
    'HandleVisibility', 'off');
plot(peak_I_time, peak_infected, 'black*', ...
    'MarkerSize', 15, 'LineWidth', 2, ...
    'HandleVisibility', 'off');

% Formatting
xlabel('Time-(days)', 'FontSize', 14, ...
    'FontWeight', 'bold');
ylabel('Number-of-Individuals', 'FontSize', 14, ...
    'FontWeight', 'bold');
title(sprintf([ 'SEIR-Epidemic-Model:-R_0= %.2f, -' ...
    '----Latent-Period= %.1f-days' ], ...
    R0, latent_period), 'FontSize', 16, ...
    'FontWeight', 'bold');
legend('Location', 'east', 'FontSize', 12);
grid on;
set(gca, 'FontSize', 12);
xlim([0 200]);
ylim([0 N]);

% Add text box with key information
annotation('textbox', [0.645, 0.63, 0.25, 0.15], ...
    'String', {sprintf('R_0= %.2f', R0), ...
        sprintf('Latent-period= %.1f-days', ...
            latent_period), ...
        sprintf('Peak-(E):- %.0f-at-T= %.1f-days', ...
            peak_exposed, peak_E_time), ...
        sprintf('Peak-(I):- %.0f-at-T= %.1f-days', ...
            peak_infected, peak_I_time), ...
        sprintf('Attack-rate:- %.1f%%', attack_rate)}, ...
    'FontSize', 10, ...
    'BackgroundColor', 'white', ...
    'EdgeColor', 'black', ...
    'LineWidth', 0.5);

hold off;

```

```
%% Save figure  
print( 'fig_SEIR_basic ', ' -dpng ', ' -r300 ' );  
fprintf( '\nFigure saved as: fig_SEIR_basic.png\n' );
```

A.4 Extended SEIR Model Simulation

```
clear; close all; clc;
```

```
%% Population
```

```
N = 1000;
```

```
%% Parameters
```

```
% Transmission rates
```

```
beta1 = 0.5;
```

```
beta2 = 0.15;
```

```
% Progression rates from E
```

```
delta1 = 1/5; % Incubation (5 days)
```

```
delta2 = 1/8;
```

```
q_E = 0.03;
```

```
% Recovery rates
```

```
gamma1 = 1/7;
```

```
gamma2 = 1/10;
```

```
gamma_Q = 1/14;
```

```
gamma_H = 1/12;
```

```
% Death rates
```

```
mu1 = 0.005;
```

```
mu2 = 0.02;
```

```
mu_Q = 0.002;
```

```
mu_H = 0.03;
```

```
% Hospitalisation
```

```
h = 0.2;
```

```
% Waning immunity
```

```
omega = 1/182;
```

```
omega_V = 1/182;
```

```
% Vaccination
```

```
v = 0.001;
```

```
%% Initial conditions
```

```
E0 = 5;
```

```
I1_0 = 3;
```

```
I2_0 = 1;
```

```

Q0 = 0;
H0 = 0;
R0_init = 0;
V0 = 0;
D0 = 0;
S0 = N - E0 - I1_0 - I2_0 - Q0 - H0 - R0_init - V0 - D0;

% Time
t_span = [0 365];
t_eval = linspace(0, 400, 2000);

%% Solve the ODE System
y0 = [S0; E0; I1_0; I2_0; Q0; H0; R0_init; V0; D0];

[t, y] = ode45(@(t,y) ode_extended_SEIR(t, y, beta1, beta2, ...
    delta1, delta2, q_E, v, omega, omega_V, ...
    gamma1, gamma2, gamma_Q, gamma_H, ...
    mu1, mu2, mu_Q, mu_H, h), t_eval, y0);

% Compartments
S = y(:,1);
E = y(:,2);
I1 = y(:,3);
I2 = y(:,4);
Q = y(:,5);
H = y(:,6);
R = y(:,7);
V = y(:,8);
D = y(:,9);

%% Find key metrics
[peak_exposed, peak_E_idx] = max(E);
peak_E_time = t(peak_E_idx);

[peak_I1, peak_I1_idx] = max(I1);
peak_I1_time = t(peak_I1_idx);

[peak_I2, peak_I2_idx] = max(I2);
peak_I2_time = t(peak_I2_idx);

final_recovered = R(end);
final_vaccinated = V(end);
total_deaths = D(end);

```

```

%% Create figure
figure('Position', [100 100 900 600]);

hold on;
plot(t, S, 'b-', 'LineWidth', 2.5, 'DisplayName', ...
     'Susceptible');
plot(t, E, 'Color', [1 0.5 0], 'LineWidth', 2.5, ...
     'DisplayName', 'Exposed');
plot(t, I1, 'r-', 'LineWidth', 2.5, 'DisplayName', ...
     'I_1');
plot(t, I2, 'm-', 'LineWidth', 2.5, 'DisplayName', ...
     'I_2');
plot(t, R, 'g-', 'LineWidth', 2.5, 'DisplayName', ...
     'Recovered');
plot(t, Q, 'c—', 'LineWidth', 1.5, 'DisplayName', ...
     'Quarantine');
plot(t, H, 'Color', [0.5 0 0.5], 'LineStyle', '—', ...
     'LineWidth', 1.5, 'DisplayName', 'Hospitalised');
plot(t, V, 'k—', 'LineWidth', 1.5, 'DisplayName', ...
     'Vaccinated');
plot(t, D, 'Color', [0.3 0.3 0.3], 'LineStyle', ':', ...
     'LineWidth', 2, 'DisplayName', 'Deceased');

% Formatting
xlabel('Time-(days)', 'FontSize', 14, 'FontWeight', ...
     'bold');
ylabel('Number-of-Individuals', 'FontSize', 14, ...
     'FontWeight', 'bold');
title('Extended-SEIR-Model:-All-Compartments', ...
     'FontSize', 16, 'FontWeight', 'bold');
legend('Location', 'northeast', 'FontSize', 9);
grid on;
set(gca, 'FontSize', 12);
xlim([0 400]);
ylim([0 N]);

hold off;

%% Save figure
print('fig_extended_SEIR', '-dpng', '-r300');
fprintf(['\nFigure-saved-as:-' ...
        '----fig_extended_SEIR.png\n']);

%% =====
% ODE function          equation

```

```

function dydt = ode_extended_SEIR(~, y, beta1, ...
    beta2, delta1, delta2, q_E, v, omega, ...
    omega_V, gamma1, gamma2, gamma_Q, ...
    gamma_H, mu1, mu2, mu_Q, mu_H, h)

S = y(1);
E = y(2);
I1 = y(3);
I2 = y(4);
Q = y(5);
H = y(6);
R = y(7);
V = y(8);
D = y(9);

N_alive = S + E + I1 + I2 + Q + H + R + V;

dS = -beta1*S*I1/N_alive - beta2*S*I2/N_alive - v*S + omega*R + omega_V*V;
dE = beta1*S*I1/N_alive + beta2*S*I2/N_alive - (delta1 + delta2 + q_E)*E;
dI1 = delta1*E - (gamma1 + mu1)*I1;
dI2 = delta2*E - (gamma2 + mu2 + h)*I2;
dQ = q_E*E - (gamma_Q + mu_Q)*Q;
dH = h*I2 - (gamma_H + mu_H)*H;
dR = gamma1*I1 + gamma2*I2 + gamma_Q*Q + gamma_H*H - omega*R;
dV = v*S - omega_V*V;
dD = mu1*I1 + mu2*I2 + mu_Q*Q + mu_H*H;

dydt = [dS; dE; dI1; dI2; dQ; dH; dR; dV; dD];
end

```

A.5 SEIR Model for COVID-19

```

function SEIR_COVID19_FirstWave()
close all; clc;

%% Population
N = 67100000;           % UK population
%% Deriving Parameters from Reported Data

% Incubation period: estimated at 5.2 days
sigma = 1 / 5.2;

% (2) Infectious period
gamma = 1 / 10;

% (3) Basic reproduction number early UK estimates placed R0
%       between 2.5 and 3.5
R0 = 3.0;
beta = R0 * gamma;

I0 = 20000;              % Estimated true infections
E0 = round(I0 * gamma / sigma); % Derived exposed count
R0_init = 0;
S0 = N - E0 - I0 - R0_init;

% Calculate derived quantities
latent_period = 1 / sigma;
infectious_period = 1 / gamma;

%% Time parameters (first wave = 219 days)
T_days = 219;
t_eval = linspace(0, T_days, 2000);

fprintf( 'SEIR-Model-for-COVID-19-(First-Wave):\n' );
fprintf( '=====\n' );
fprintf( 'Population-(N):-%d\n', N);
fprintf( 'Transmission-rate-(beta):-%.4f-per-day\n', beta);
fprintf( 'Latent-period-(1/sigma):-%.1f-days\n', latent_period);
fprintf( 'Infectious-period-(1/gamma):-%.1f-days\n', infectious_period);
fprintf( 'Progression-rate-(sigma):-%.4f-per-day\n', sigma);
fprintf( 'Recovery-rate-(gamma):-%.2f-per-day\n', gamma);
fprintf( 'Basic-reproduction-number-(R0):-%.2f\n', R0);

```

```

fprintf( 'Derived initial conditions:\n' );
fprintf( 'S(0) = %d\n', S0 );
fprintf( 'E(0) = %d (from E/I = gamma/sigma)\n', E0 );
fprintf( 'I(0) = %d (estimated true infections)\n', I0 );
fprintf( 'R(0) = %d\n', R0_init );
fprintf( '=====\n\n' );

%% Solve the ODE System
y0 = [S0; E0; I0; R0_init];

[t, y] = ode45(@(t,y) ode_SEIR(t, y, beta, sigma, gamma, ...
    N), t_eval, y0);

% Compartments
S = y(:,1); % Susceptible
E = y(:,2); % Exposed
I = y(:,3); % Infected
R = y(:,4); % Recovered

%% Find key metrics
[peak_exposed, peak_E_idx] = max(E);
peak_E_time = t(peak_E_idx);

[peak_infected, peak_I_idx] = max(I);
peak_I_time = t(peak_I_idx);

final_recovered = R(end);
attack_rate = (final_recovered / N) * 100;

fprintf( 'Epidemic Outcomes (First Wave):\n' );
fprintf( '=====\n' );
fprintf( 'Peak-exposed: %0.0f individuals\n', peak_exposed );
fprintf( 'Time-to-peak-exposed: %0.1f days\n', peak_E_time );
fprintf( 'Peak-infected: %0.0f individuals\n', peak_infected );
fprintf( 'Time-to-peak-infected: %0.1f days\n', peak_I_time );
fprintf( 'Time-delay (E-peak-to-I-peak): %0.1f days\n', ...
    peak_I_time - peak_E_time );
fprintf( 'Final-attack-rate: %0.1f%%\n', attack_rate );
fprintf( 'R0 = beta/gamma = %0.2f\n', R0 );
fprintf( '=====\n' );

%% Create figure
figure('Position', [100 100 900 600]);

hold on;

```

```

plot(t, S, 'b-', 'LineWidth', 2.5, 'DisplayName', ...
    'Susceptible-(S)');
plot(t, E, 'm-', 'LineWidth', 2.5, 'DisplayName', ...
    'Exposed-(E)');
plot(t, I, 'r-', 'LineWidth', 2.5, 'DisplayName', ...
    'Infected-(I)');
plot(t, R, 'g-', 'LineWidth', 2.5, 'DisplayName', ...
    'Recovered-(R)');

% Mark peak exposed and infected
plot(peak_E_time, peak_exposed, 'black*', ...
    'MarkerSize', 15, 'LineWidth', 2, ...
    'HandleVisibility', 'off');
plot(peak_I_time, peak_infected, 'black*', ...
    'MarkerSize', 15, 'LineWidth', 2, ...
    'HandleVisibility', 'off');

% Formatting
xlabel('Time-(days)', 'FontSize', 14, ...
    'FontWeight', 'bold');
ylabel('Number-of-Individuals', 'FontSize', 14, ...
    'FontWeight', 'bold');
title(sprintf(['SEIR-Model:-COVID-19-First-Wave-' ...
    '(R_0= %.2f, -Latent-Period = %.1f-days)'], ...
    R0, latent_period), 'FontSize', 16, ...
    'FontWeight', 'bold');
legend('Location', 'east', 'FontSize', 12);
grid on;
set(gca, 'FontSize', 12);
xlim([0 T_days]);
ylim([0 N]);

% Add text box with key information
annotation('textbox', [0.594, 0.55, 0.3, 0.22], ...
    'String', {sprintf('R_0= %.2f', R0), ...
        sprintf('Latent-period = %.1f-days', ...
            latent_period), ...
        sprintf('Infectious-period = %.1f-days', ...
            infectious_period), ...
        sprintf('E(0) = %d, -I(0) = %d', E0, I0), ...
        sprintf('Peak-(E): - %.0f-at-T= %.1f-days', ...
            peak_exposed, peak_E_time), ...
        sprintf('Peak-(I): - %.0f-at-T= %.1f-days', ...
            peak_infected, peak_I_time), ...
        sprintf('Attack-rate: - %.1f%%', attack_rate)}, ...
    'FontSize', 10, ...

```

```

        'BackgroundColor', 'white', ...
        'EdgeColor', 'black', ...
        'LineWidth', 0.5);
hold off;

%% Save figure
print('fig_SEIR_COVID19_first_wave', 'dpng', 'r300');
fprintf('\nFigure saved as: fig_SEIR_COVID19_first_wave.png\n');

end

%% ODE function
function dydt = ode_SEIR(~, y, beta, sigma, gamma, N)
    S = y(1);
    E = y(2);
    I = y(3);
    R = y(4);

    dS = -beta * S * I / N;
    dE = beta * S * I / N - sigma * E;
    dI = sigma * E - gamma * I;
    dR = gamma * I;

    dydt = [dS; dE; dI; dR];
end

```

A.6 Intervention Scenarios

```
clear; clc; close all;
```

```
%% UK second wave parameters
```

```
N = 66739900;      % UK population  
T_days = 250;      % approx second wave
```

```
% — Epidemiological parameters
```

```
% Transmission rates
```

```
beta1 = 0.35;  
beta2 = 0.12;
```

```
% Progression rates from E
```

```
delta1 = 1/5.2;      %incubation (5.2 days)  
delta2 = 1/8.0;      % slower, identified individuals
```

```
% Recovery rates
```

```
gamma1 = 1/7;  
gamma2 = 1/10;  
gamma_Q = 1/14;  
gamma_H = 1/12;
```

```
% Death rates
```

```
mu1 = 0.005;  
mu2 = 0.02;  
mu_Q = 0.002;  
mu_H = 0.03;
```

```
% Hospitalisation
```

```
h_rate = 0.3;
```

```
% Waning immunity
```

```
omega = 1/180;      % 6 months  
omega_V = 1/180;
```

```
% — Control parameters —
```

```
v_vax = 0.003;      % approx 200k/day at peak offic of
```

```

                                % national stats

q_E_val = 0.05;                % quarantine rate

%% Initial conditions (fractions of N)
I1_0 = 3000 / N;
E_0  = 2 * I1_0;
I2_0 = I1_0 * 0.15;
H_0  = 500 / N;
Q_0  = 0;
R_0  = 0.05;                    % 5% immune from first wave
V_0  = 0;
D_0  = 41000 / N;               % cumulative deaths from first wave
S_0  = 1 - E_0 - I1_0 - I2_0 - Q_0 - H_0 - R_0 - V_0 - D_0;

y0 = [S_0, E_0, I1_0, I2_0, Q_0, H_0, R_0, V_0, D_0];

tspan = [0 T_days];

%% Run three scenarios
% No intervention (v=0, q-E=0)
params1 = [beta1, beta2, delta1, delta2, 0, 0,
           omega, omega_V, gamma1, gamma2, gamma_Q,
           gamma_H, mu1, mu2, mu_Q, mu_H, h_rate];
[t1, y1] = ode45(@(t,y) extended_model(t, y, ...
           params1), tspan, y0);

% Vaccination only (v>0, q-E=0)
params2 = [beta1, beta2, delta1, delta2, 0,
           v_vax, omega, omega_V, ...
           gamma1, gamma2, gamma_Q,
           gamma_H, mu1, mu2, mu_Q, mu_H,
           h_rate];
[t2, y2] = ode45(@(t,y) extended_model ...
           (t, y, params2), tspan, y0);

% v>0, q-E>0
params3 = [beta1, beta2, delta1, delta2,
           q_E_val, v_vax, omega, omega_V, ...
           gamma1, gamma2, gamma_Q,
           gamma_H, mu1, mu2, mu_Q, mu_H,
           h_rate];

```

```

[t3, y3] = ode45(@(t,y) extended_model ...
    (t, y, params3), tspan, y0);

%% Figure: stacked plots
figure('Position', [100 100 600 1000]);

% (a) Infected population I1 + I2
subplot(2,1,1);
plot(t1, (y1(:,3) + y1(:,4)) * N, 'r-', ...
    'LineWidth', 2); hold on;
plot(t2, (y2(:,3) + y2(:,4)) * N, ...
    'Color', [1 0.5 0], 'LineStyle', ...
    '—', 'LineWidth', 2);
plot(t3, (y3(:,3) + y3(:,4)) * N, ...
    'g-', 'LineWidth', 2);
xlabel('Time-(Days)', 'FontSize', 12);
ylabel('Active-Infections-(I_1+I_2)', ...
    'FontSize', 12);
title(['Infected-Population-' ...
    '---Under-Different-Scenarios'], ...
    'FontSize', 13);
legend('No-Intervention', ...
    'Vaccination-Only', 'Combined-(Vax+Q)', ...
    'Location', 'northwest', 'FontSize', 10);
grid on; grid minor;

% total deaths
subplot(2,1,2);
plot(t1, (y1(:,9) - y1(1,9)) * N, 'r-', ...
    'LineWidth', 2); hold on;
plot(t2, (y2(:,9) - y2(1,9)) * N, 'Color', [1 0.5 0], ...
    'LineStyle', '—', 'LineWidth', 2);
plot(t3, (y3(:,9) - y3(1,9)) * N, 'g-', ...
    'LineWidth', 2);
xlabel('Time-(Days)', 'FontSize', 12);
ylabel('Cumulative-Deaths-(Second-Wave)', ...
    'FontSize', 12);
title('Total-Deaths-Under-Different-Scenarios', ...
    'FontSize', 13);
legend('No-Intervention', 'Vaccination-Only', ...
    'Combined-(Vax+Q)', ...
    'Location', 'northwest', 'FontSize', 10);
grid on; grid minor;

```

```
%% Extended model ODE function
```

```
function dydt = extended_model(~, y, p)
    S = y(1); E = y(2); I1 = y(3); I2 = y(4);
    Q = y(5); H = y(6); R = y(7); V = y(8); D = y(9);

    b1 = p(1); b2 = p(2);
    d1 = p(3); d2 = p(4); qE = p(5);
    v = p(6); om = p(7); omV = p(8);
    g1 = p(9); g2 = p(10); gQ = p(11); gH = p(12);
    m1 = p(13); m2 = p(14); mQ = p(15); mH = p(16);
    hr = p(17);

    N_alive = S + E + I1 + I2 + Q + H + R + V;

    dS = -b1*S*I1/N_alive - b2*S*I2/N_alive - v*S + om*R + omV*V;
    dE = b1*S*I1/N_alive + b2*S*I2/N_alive - (d1 + d2 + qE)*E;
    dI1 = d1*E - (g1 + m1)*I1;
    dI2 = d2*E - (g2 + m2 + hr)*I2;
    dQ = qE*E - (gQ + mQ)*Q;
    dH = hr*I2 - (gH + mH)*H;
    dR = g1*I1 + g2*I2 + gQ*Q + gH*H - om*R;
    dV = v*S - omV*V;
    dD = m1*I1 + m2*I2 + mQ*Q + mH*H;

    dydt = [dS; dE; dI1; dI2; dQ; dH; dR; dV; dD];
end
```

A.7 Sensitivity Analysis

```
clear; clc; close all;
```

```
%% UK second wave parameters
```

```
N = 66739900;
```

```
T_days = 250;
```

```
%% Initial parameters
```

```
baseline = [0.35, % beta1  
            0.12, % beta2  
            1/5.2, % delta1  
            1/8.0, % delta2  
            0.05, % q-E  
            0.003, % v  
            1/180, % omega  
            1/180, % omega_V  
            1/7, % gamma1  
            1/10, % gamma2  
            1/14, % gamma_Q  
            1/12, % gamma_H  
            0.005, % mu1  
            0.02, % mu2  
            0.002, % mu_Q  
            0.03, % mu_H  
            0.3]; % h
```

```
% Varying Parameters
```

```
param_names = {'\beta_1', '\delta_1', '\gamma_1', 'v', 'q-E', 'h'};  
param_indices = [1, 3, 9, 6, 5, 17];
```

```
perturbation = 0.10; % +/- 10%
```

```
%% Initial conditions (fractions of N)
```

```
I1_0 = 3000 / N;
```

```
E_0 = 2 * I1_0;
```

```
I2_0 = I1_0 * 0.15;
```

```
H_0 = 500 / N;
```

```
Q_0 = 0;
```

```
R_0 = 0.05;
```

```
V_0 = 0;
```

```
D_0 = 41000 / N;
```

```
S_0 = 1 - E_0 - I1_0 - I2_0 - Q_0 - H_0 - R_0 - V_0 - D_0;
```

```

y0 = [S_0, E_0, I1_0, I2_0, Q_0, H_0, R_0, V_0, D_0];
tspan = [0 T_days];

%% Base values
baseline_cum = compute_cumulative_infections(baseline, y0, tspan, N);

%% OAT Sensitivity Analysis
n_params = length(param_indices);
low_pct = zeros(n_params, 1);
high_pct = zeros(n_params, 1);

fprintf('—— Results ——\n');
for k = 1:n_params
    idx = param_indices(k);

    % Low perturbation (-10%)
    p_low = baseline;
    p_low(idx) = baseline(idx) * (1 - perturbation);
    cum_low = compute_cumulative_infections(p_low, y0, tspan, N);

    % High perturbation (+10%)
    p_high = baseline;
    p_high(idx) = baseline(idx) * (1 + perturbation);
    cum_high = compute_cumulative_infections(p_high, y0, tspan, N);

    low_pct(k) = (cum_low - baseline_cum) / baseline_cum * 100;
    high_pct(k) = (cum_high - baseline_cum) / baseline_cum * 100;

    fprintf('%8s: -10%%->-%12.0f-(%+1f%%)- | -+10%%->-%12.0f-(%+1f%%)\n', ...
        param_names{k}, cum_low, low_pct(k), cum_high, high_pct(k));
end

%% Tornado Plot
% Sort by total swing
total_swing = abs(high_pct - low_pct);
[~, sort_idx] = sort(total_swing, 'ascend');

sorted_names = param_names(sort_idx);
sorted_low = low_pct(sort_idx);
sorted_high = high_pct(sort_idx);

figure('Position', [100 100 1000 600]);
y_pos = 1:n_params;

for k = 1:n_params

```

```

    % -10% perturbation
    barh(y_pos(k), sorted_low(k), 0.55, 'FaceColor', [1 0 0], ...
        'FaceAlpha', 0.85, 'EdgeColor', 'w');
    hold on;
    % +10% perturbation
    barh(y_pos(k), sorted_high(k), 0.55, 'FaceColor', [0 0 1], ...
        'FaceAlpha', 0.85, 'EdgeColor', 'w');
end

% Formatting
set(gca, 'YTick', y_pos, 'YTickLabel', sorted_names, 'FontSize', 13);
xlabel('Percentage Change in Total Infections (%)', 'FontSize', 12);
title('Sensitivity of Total Infections to \pm 10% Parameter Variation', ...
    'FontSize', 13, 'FontWeight', 'bold');
xline(0, 'k-', 'LineWidth', 1);
grid on;

%% Helper function: compute cumulative infections

function cum_inf = compute_cumulative_infections(params, y0, tspan, N)
[t, y] = ode45(@(t,y) extended_model(t, y, params), tspan, y0);

% Cumulative infections = integral of (delta1 + delta2 + q_E) * E(t)
d1 = params(3);
d2 = params(4);
qE = params(5);
outflow_E = (d1 + d2 + qE) .* y(:,2);
cum_inf = trapz(t, outflow_E) * N;
end

%% Extended model ODE function      equation (1)

function dydt = extended_model(~, y, p)
S = y(1); E = y(2); I1 = y(3); I2 = y(4);
Q = y(5); H = y(6); R = y(7); V = y(8); D = y(9);

b1 = p(1); b2 = p(2);
d1 = p(3); d2 = p(4); qE = p(5);
v = p(6); om = p(7); omV = p(8);
g1 = p(9); g2 = p(10); gQ = p(11); gH = p(12);
m1 = p(13); m2 = p(14); mQ = p(15); mH = p(16);
hr = p(17);

N_alive = S + E + I1 + I2 + Q + H + R + V;

```

```

dS  = -b1*S*I1/N_alive - b2*S*I2/N_alive - v*S + om*R + omV*V;
dE  =  b1*S*I1/N_alive + b2*S*I2/N_alive - (d1 + d2 + qE)*E;
dI1 =  d1*E - (g1 + m1)*I1;
dI2 =  d2*E - (g2 + m2 + hr)*I2;
dQ  =  qE*E - (gQ + mQ)*Q;
dH  =  hr*I2 - (gH + mH)*H;
dR  =  g1*I1 + g2*I2 + gQ*Q + gH*H - om*R;
dV  =  v*S - omV*V;
dD  =  m1*I1 + m2*I2 + mQ*Q + mH*H;

dydt = [dS; dE; dI1; dI2; dQ; dH; dR; dV; dD];
end

```

A.8 Effect of vaccination on SEIR epidemics

```

clear; close all; clc;

%% Model Parameters
N = 1000;
gamma = 1/10;           % 10 day infectious period
delta = 1/5;           % 5 day incubation period
R0 = 2.5;
beta = R0 * gamma;     % Transmission rate

T = 365;                % Simulation (days/year)
t_eval = linspace(0, T, 1000);

%% SEIR Model (no vaccination)
function dydt = SEIR_equations(t, y, beta, delta, gamma, N)
    S = y(1);
    E = y(2);
    I = y(3);
    R = y(4);

    dS = -beta * S * I / N;
    dE = beta * S * I / N - delta * E;
    dI = delta * E - gamma * I;
    dR = gamma * I;

    dydt = [dS; dE; dI; dR];
end

%% Pre-Vaccination (reduce S(0) before epidemic starts)

vaccination_levels = [0, 0.20, 0.40, 0.60];
colors = {[1 0 0], [1 0.5 0], [0 0.6 0], [0 0 1]};
linestyles = {'-', '-', '-', '-'};

figure('Position', [100 100 800 500]);
hold on;
for k = 1:length(vaccination_levels)
    v = vaccination_levels(k);

    % Initial conditions: move fraction v from S to R
    S0 = N * (1 - v) - 1;
    E0 = 0;
    I0 = 1;

```

```

R0_init = N * v;

y0 = [S0; E0; I0; R0_init];

[t, y] = ode45(@(t,y) SEIR_equations(t, y, beta, delta, gamma, N), t_eval, y0);

I_curve = y(:,3);
[peak_I, peak_idx] = max(I_curve);
peak_t = t(peak_idx);

Rv = (1 - v) * R0;
label = sprintf('v=%0.2f (R_v=%0.2f)', v, Rv);

plot(t, I_curve, 'Color', colors{k}, 'LineStyle', linestyles{k}, ...
      'LineWidth', 3, 'DisplayName', label);

% Mark peak with asterisk
if peak_I > 5
    plot(peak_t, peak_I, '*', 'Color', colors{k}, 'MarkerSize', 15, ...
          'LineWidth', 2, 'HandleVisibility', 'off');
end

end

% Formatt
xlabel('Time (days)', 'FontSize', 14, 'FontWeight', 'bold');
ylabel('Infected Individuals', 'FontSize', 14, 'FontWeight', 'bold');
title(sprintf('Effect of Vaccination on SEIR Epidemic Dynamics (R_0=%0.2f)', R0), 'FontSize', 16, 'FontWeight', 'bold');
legend('Location', 'northeast', 'FontSize', 11);
grid on;
set(gca, 'FontSize', 12, 'LineWidth', 1.5);
xlim([0 T]);
ylim([0 165]);
hold off;

```